A&A TRANSFER

Crane Lift Plan

for

Rosendin Electric
Stack NVA05A
Manassas, VA

Hoist & Set Transformer

Table of Contents

44175 Lavin Lane, Chantilly, VA 20152 Office: (703) 471-8338 Fax: (703) 327-6634 www.AARigging.com

1 – Lift Plan Overview/AHA/JHA

Lift Plan Overview

General Information

Date Created:	11/7/2024	Lift Date:	TBD		
Customer:	Rosendin Electric				
Job Name:	Stack NVA05A				
Street Address:	9101 Freedom Center Drive				
City:	Manassas	State:	VA	Zip:	20110
Scope of Work:	Hoist & Set Transformer				

Crane Information

Crane Owner:	WO Grubb	Type:	Hydraulic Truck Crane
Manufacturer:	Liebherr	Capacity (tons):	235
Model:	LTM 1200-5.1	Inspection Exp. Date:	See Attached
Serial #:	See Attached		

Load Information

Description:	Transformer
Dimensions:	8' L x 8' W x 8' H
Weight:	16,500 lbs

Lift Information

Radius:	95'	Net Load Weight:	17,717
Swing Range:	360	Crane Capacity:	24,200
Configuration:	Main Boom	% of Load Chart:	73.00

Operator Information

	Name	Employer
Primary Operator:	Brandon Ashley	WO Grubb
Back Up Operator:	TBD	WO Grubb

A&A TRANSFER

CRANE BELOW THE HOOK PRE-LIFT CHECKLIST

*Complete fillable format in Safety Docs Template

Yes/No	
YES / NO ☐ ☐	Is the appropriate rigging plan in hand/on site?
YES / NO ☐ ☐	Is the operator onsite the same operator shown in the plan?
YES / NO ☐ ☐	Is the crane configuration on site the same as the configuration shown in the plan?
YES / NO ☐ ☐	Is the rigging configuration (below the hook) onsite the same configuration as shown in the plan?
YES / NO ☐ ☐	Do factory provided lifting points show any visible damage? (If "Yes" explain)
YES / NO ☐ ☐	Does rigging configuration match factory recommendation (if provided)?
YES / NO ☐ ☐	Do site conditions allow the delivery trucks to stage with-In lift radius approved in plan? (If "No" explain)
YES / NO ☐ ☐	Does the final set method onsite match the one approved in the plan? (Gantry, anchor plates, chain falls, forklift) (If "No" contact Project Manager for review)
YES / NO ☐ ☐	Is there a submittal data in the plan or onsite for the equipment being picked? (showing weights and dimensions)
YES / NO ☐ ☐	Do the weights of the items being picked appear to match the weight identified in the approved plan /submittals?
YES / NO ☐ ☐	Has lift test been performed with first piece of equipment being hosted?

NAME:	SIGNATURE:	DATE:
OPERATOR:		
Rigging Supervisor:		
Signal Person:		
Client/Third Party:		

Activity Hazard Analysis (AHA)

Activity/Work Task:Hoist & SetTransformer	Overall Risk Assessment Code (RAC) (Use highest code)					M
Project Location: Manassas, VA	**Risk Assessment Code (RAC) Matrix**					
Contract Number: N/A	**Severity**	**Probability**				
Date Prepared: 11/7/2024		Frequent	Likely	Occasional	Seldom	Unlikely
Prepared by (Name/Title): Hugo Moreno / Estimator	Catastrophic	E	E	H	H	M
	Critical	E	H	H	M	L
Reviewed by (Name/Title): Dewey Snavely / VP Field Ops	Marginal	H	M	M	L	L
	Negligible	M	L	L	L	L

Notes: (Field Notes, Review Comments, etc.)

- All involved personnel will use proper PPE at all Times: Hard hats, leather gloves, steel toe boots, safety glasses, and safety reflective vest.
- All involved personnel will use the 10 foot rule: While handling moving cable/hook personnel must keep hands and or body at least 10 feet away or sheave must be 10 feet in height.

Step 1: Review each **"Hazard"** with identified safety **"Controls"** and determine RAC (See above)

	RAC Chart
"Probability" is the likelihood to cause an incident, near miss, or accident and identified as: Frequent, Likely, Occasional, Seldom or Unlikely.	
"Severity" is the outcome/degree if an incident, near miss, or accident did occur and identified as: Catastrophic, Critical, Marginal, or Negligible	**E = Extremely High Risk** **H = High Risk**
Step 2: Identify the RAC (Probability/Severity) as E, H, M, or L for each "Hazard" on AHA. Annotate the overall highest RAC at the top of AHA.	**M = Moderate Risk** **L = Low Risk**

Principal/Job Steps	Hazards	Controls	RAC
Logistics and preparation for beginning work.	• permits required, training, emergency egress, order of work.	• Make sure all necessary permits have been obtained and filled out daily prior to work beginning. Make sure all personnel have been trained in the tasks that will be performed. Make sure everyone involved knows the evacuation routes and what to do in case of an emergency. Make sure that all workers know the plan for the task and are on agreement with the plan for the day. Make sure everyone has all standard PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest) and has additional safety equipment	L
Monitor Severe Weather	• thunder / Lightning Storms • wind • tornados / Hurricanes	• Designate and establish an area of retreat and communicate this location to all involved personnel • If warnings have been issued or if there are indications of severe weather in the vicinity, monitor weather using NOAA weather stations • If it is determined that severe weather will impact the project, shut down operations and move all personnel to the 'area of retreat' until it is determined safe • Establish that crane operations will be suspended in sustained winds exceeding 20 mph. Suspension of operations may occur at less than 20 mph if the operator or person in charge determines that continued operations will create a hazard.	L

Prepare Site	• slips and trips. Poor housekeeping, and particles in eyes	• Inspect and clear site for delivery truck. Ensure that there are no holes or debris that will hinder the equipment. Wear all needed PPE. • Ensure proper clearances from power lines and follow 29 CFR 1926.1408 Table A. Clearances listed below	L
Drive crane into location	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).Use proper lifting techniques.	L
Back assist tractor trailer up to crane	• pinch points, slips and trips, and struck by/caught in between	• Inspect and clear site for delivery truck. Ensure that there are no holes or debris that will hinder the equipment. Use back-up alarm and or spotters with flag or paddle to back truck into position. Wear all needed PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).	L
Remove hand pads from trailer and put in desired location for crane	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).Use proper lifting techniques.	L
Extend outriggers half way and level crane on hand pads	• pinch points	• Be aware of hand and body placements.	L
Attach chain bridle to hook and inspect	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).Use proper lifting techniques. Inspect all equipment per manufactures requirements and inspect all rigging equipment.	L
Attach chain bridle to large mats on trailer and locate in desired location	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques. Inspect all equipment per manufactures requirements and inspect all rigging equipment.	M
Reposition crane onto large mats and level crane	• pinch points	• Be aware of hand and body placements.	L
Set barricade up using cones and danger tape to protect swing radius of crane.	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques. Cones and tape to be set beyond maximum swing radius of counterweights.	L

Detach chain bridle and attach counterweight cables/slings to hook and inspect	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques. Inspect all equipment per manufactures requirements and inspect all rigging equipment.	L
Attach counterweight cables/slings to counterweight and place on deck of crane (repeat for all counterweights)	• pinch points, muscle strains, hand injuries, head injuries, slips and trips, back injuries, falls	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques. Inspect all equipment per manufactures requirements and inspect all rigging equipment. 100% tie off if over 6 feet (Ladder, harness, lanyard, yoyo)	M
Swing crane into position and bring counterweights up to working position	• pinch points	• Be aware of hand and body placements.	L
Using either the flatbed 4' tall trailer or an 8' step ladder, whichever is more accessible, reeve cable through sheaves and to load block	• pinch points • muscle strains • hand injuries • head injuries • slips and trips • back injuries, falls	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques. 100% tie off if over 6 feet. 3 points of contact when any work is to be done off a ladder at no greater than 6'.	L
Attach anti-two block device to cable and connect to switch on jib and attach wind meter to crane.	• pinch points • muscle strains • hand injuries • head injuries • slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques. No work to be done past 23fps or 16 mph.	L
Re-level crane and swing boom into safe lifting position	• pinch points	• Be aware of hand and body placements.	L
Perform inspection of all components to be in safe operating conditions	• pinch points • head injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).	L
Attach rigging to load hook and inspect	• pinch points • muscle strains • hand / heady injuries • slips and trips, back injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Use proper lifting techniques.	L
Equipment and Rigging Inspection	• faulty equipment	• Inspect all equipment per manufactures requirements and inspect all rigging equipment.	L
Attach rigging to material to be lifted.	• pinch points • muscle strains • hand injuries	• Be aware of hand and body placement. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).Use proper lifting techniques.	L

Attach rigging to crane hook.	• back injuries • struck by • hand / head injuries	• Be aware of all crane movements. Wear proper PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). Pay attention to task.	L
Perform lifts to desired location.	• crushing • struck by • pinch points • overhead hazards • electrocution • operator error	• Only qualified operator will run the crane and signalman will be designated. • Do not stand or place any body parts under suspended loads. • Stay clear of load path of travel and move clear of the path to avoid getting struck by load. Stay clear of counterweight swing area. Clear landing area of non-essential personnel. Be aware of crane movement. • Use tag lines to control the load • Presume all power lines to be energized unless utility owner/operator confirms that power line has been and continues to be de-energized and visibly grounded at the work site. (1926.1408) • Maintain clearance of all crane & load parts from power lines if applicable. Up to 50kV=10ft ; 50-200kV=15ft ; 200-350kV=20ft ; 350-500kV=25ft ; 500-750kV=35 ft ; 750-1000kV=45ft • Over1,000kV (as established by utility owner/operator or registered professional engineer who is qualified person with respect to electrical power transmission and distribution). • Wear all required PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest). • Crane signals will be done with 2-way radios when hoisting to the roof and 2-way radios or hand signals on the ground.	M
Break down work area and daily clean up.	• housekeeping • particles in eye • struck by • slips and trips	• Use flaggers and spotter when moving mobile equipment, clean up all debris and dispose of in appropriate container. Use all required PPE (hard hat, steel toe boots, safety glasses, leather gloves, reflective vest).	M

Equipment to be Used	**Training Requirements/Competent or Qualified Personnel name(s)**	**Inspection Requirements**

1. Liebherr LTM 1200-5.1 (235 Ton) 2. Rigging equipment "see attached"	• Use of Qualified Crane Operator (1926.1427) • Use of Qualified Riggers (1926.251) **Competent Person supervising the work will be: (TBD)** All Rigging equipment used by qualified personnel and is in good working order.	Cranes and Derricks in Construction (1926.1412) Crane Annual/Daily/Periodic Inspections (1926.1412(d)(1) Power Line Safety (1926.1408) Inspection of Ground Conditions (1926.1412(d)(1)(ix) Inspection of Rigging Equipment (1926.251(a) Daily Rigging Equipment Inspection (1926.251) Daily Rigging Inspection by competent person. All tags and capacities are clearly labeled.

THE NAME THAT CARRIES WEIGHT

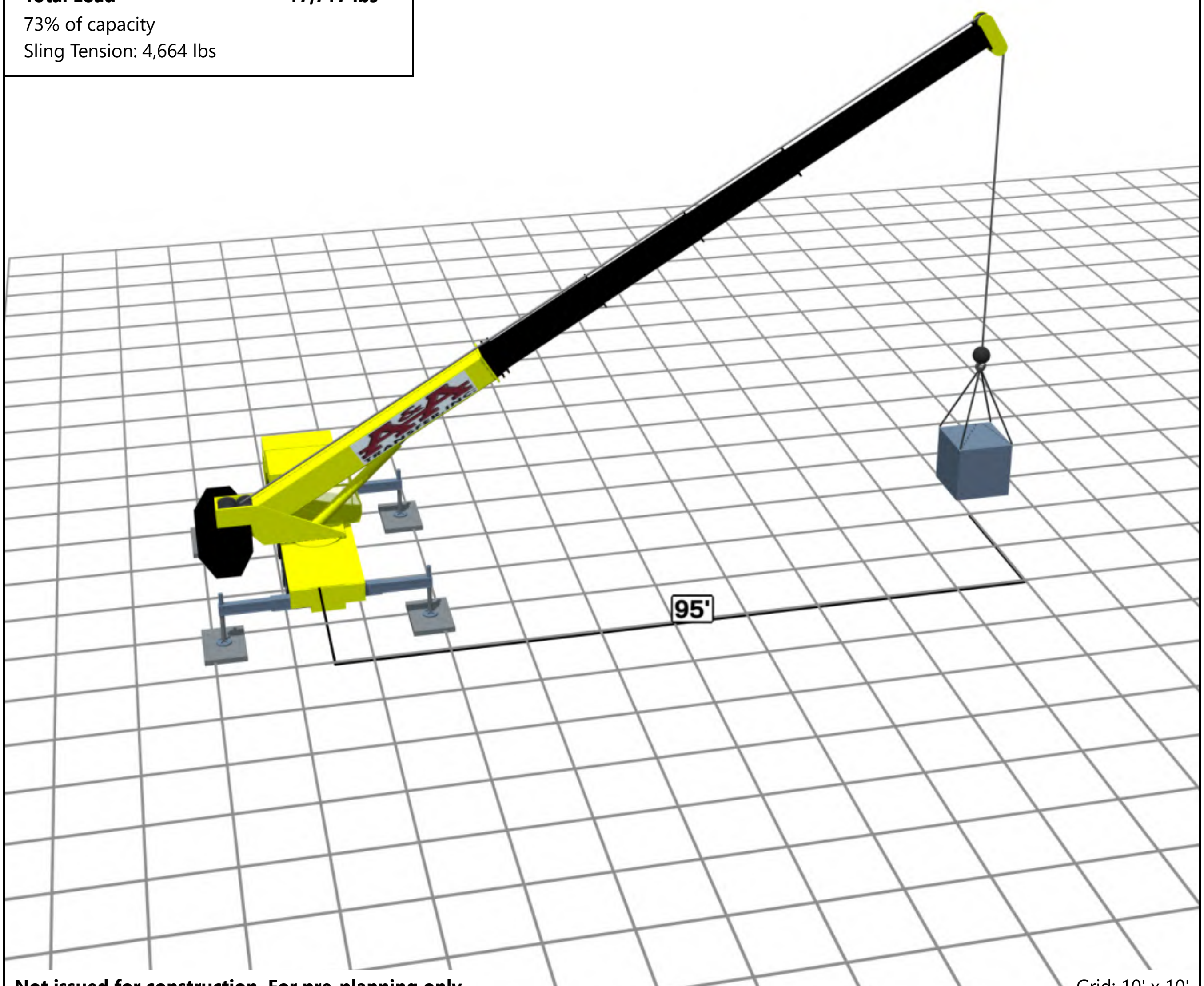
2 – Lift Image(s)

Crane
Liebherr LTM1200-5.1
117' Telescopic Boom (T) at 31.3°
Base: 100% Outriggers (29' x 27')
Counterweight: 70,600 lbs
95' Lift Radius (360°)
Crane Capacity at 95' = 24,200 lbs

Load

Load Line	86 lbs
Block	1,102 lbs
Hook	0 lbs
Sling (4)	24 lbs
Crosby Shackle G-209 5/8" (4)	5 lbs
Total Rigging Weight	**1,217 lbs**
Load	16,500 lbs
Total Load	**17,717 lbs**

73% of capacity
Sling Tension: 4,664 lbs



Not issued for construction. For pre-planning only.

Grid: 10' x 10'

Title	Lift Plan
Project	Stack NVA05A - Transformer (235T)
Customer	Rosendin Electric
Description	Hoist & Set Transformer
Drawn By	Hugo Moreno 11/7/2024

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COLOR LEGEND

LOOP A	(blue)
LOOP B	(salmon)
LOOP C	(green)
LOOP D	(red)
LOOP E	(cyan)
LOOP F	(purple)
MV TIE	(grey)

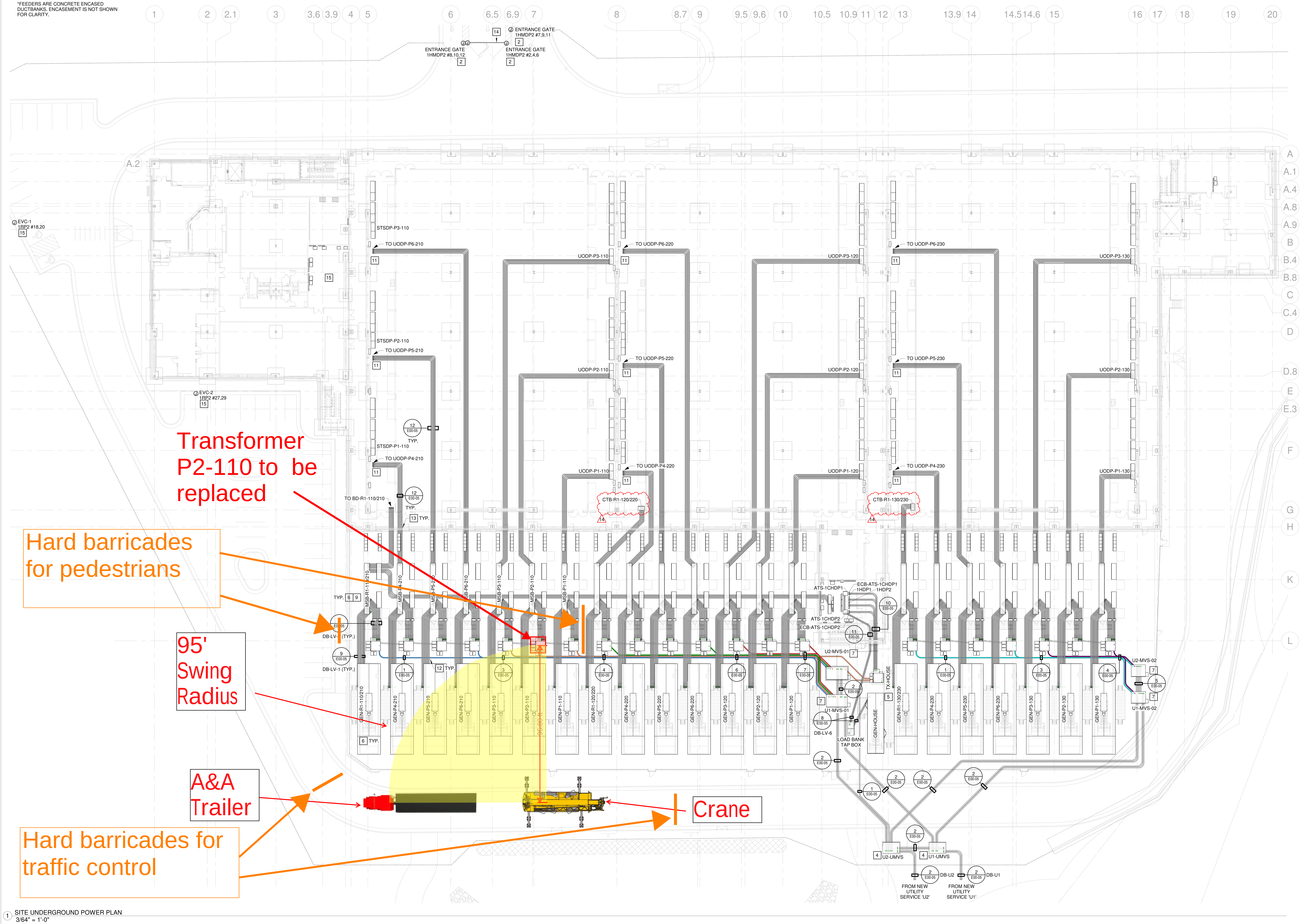
*FEEDERS ARE CONCRETE ENCASED DUCTBANKS. ENCASEMENT IS NOT SHOWN FOR CLARITY.

KEYED NOTES

1. NOT USED.
2. 480V GATE OPERATOR POWER CONNECTION. EC TO PROVIDE NEMA 3R DISCONNECT. REFER TO TELECOM DRAWINGS FOR SECURITY AND CONTROL DETAILS.
3. NOT USED.
4. PROVIDE CONDUIT FOR UMVS EPMS FROM NEAREST IDF WITHIN BUILDING TO UMVS LOCATION.
5. EC TO PROVIDE ALL CONDUIT IN THIS PHASE AS REQUIRED FOR CONNECTION FROM BUILDING TO HOUSE GEN AND HOUSE TX. EC TO COORDINATE ALL REQUIRED CONDUITS AND CIRCUITS INCLUDING, BUT NOT EXCLUSIVELY:
   - **GENERATOR COMMS AND CONTROLS:** UNDERGROUND FROM ATS-1CHDP1 TO GEN-HOUSE. UNDERGROUND FROM ATS-1CHDP2 TO GEN-HOUSE.
   - **GENERATOR START/STOP SIGNAL:** UNDERGROUND FROM ATS-1CHDP1 TO GEN-HOUSE. UNDERGROUND FROM ATS-1CHDP2 TO GEN-HOUSE.
   - **GENERATOR EPMS/BMS:** UNDERGROUND FROM BUILDING NETWORK SWITCH TO GEN-HOUSE.
   - **GENERATOR SHORE POWER:** UNDERGROUND FROM 1CHDP1 TO GEN-HOUSE.
6. REFER TO E02 SERIES DRAWINGS FOR ADDITIONAL INTERCONNECTION CONDUITS BETWEEN TRANSFORMERS, GENERATORS, AND MERS.
7. PROVIDE 1"C UNDERGROUND FROM HOUSE ELEC 139 TO MVS LOCATION FOR 120V UPS CONTROL CIRCUIT.
8. NOT USED.
9. CABLE VAULTS UNDER ENTIRE MER ARE NOT ACCEPTABLE.
10. NOT USED.
11. EXTEND CONDUITS THROUGH FIRST FLOOR GALLERY TO SECOND FLOOR EQUIPMENT LOCATIONS. COORDINATE EXACT LOCATIONS WITH ALL DISCIPLINES PRIOR TO ROUGH-IN.
12. ROUTE MV CONDUITS A MINIMUM OF 2'-0" UNDER 480V CONDUITS WHERE CONDUITS CROSS. E.C. TO VERIFY DEPTHS WITH HEAT CALCULATIONS PRIOR TO INSTALLATION.
13. REFER TO STRUCTURAL DRAWINGS FOR DUCTBANK AND FOUNDATION DETAILS.
14. ROUTE 1"C BETWEEN MAIN CONNECTION BOX AND SECONDARY MOTOR CONNECTION.
15. PROVIDE UNDERGROUND CONDUIT TO HANDHOLE ADJACENT TO CAR CHARGER. HANDHOLE IS TO BE SIZED TO ACCOMMODATE UP TO (6) DUAL OUTPUT CAR CHARGERS (40A/2P)

GENERAL NOTES

1. REFER TO PROJECT GENERAL NOTES ON E00-01 FOR ADDITIONAL INFORMATION.
2. EXACT LOCATIONS OF ALL POWER DUCTBANKS, PADS, AND MANHOLES SHALL BE AS INDICATED ON CIVIL PLANS.
3. SEE CIVIL DRAWINGS FOR ADDITIONAL ROUTING DETAILS.
4. COORDINATE EXACT ROUTING AND BURIAL DEPTH OF POWER DUCTBANKS AND CONDUITS TO AVOID CONFLICTS WITH OTHER UTILITIES.
5. UNDERGROUND DUCTBANK DESIGN SHALL BE COORDINATED, COMPLETED AND CONFIRMED BY THE E.C. NEHER-MCGRATH CALCULATIONS SHALL BE PERFORMED BY THE E.C. IN ACCORDANCE WITH SPECIFICATIONS.
6. REFER TO SPECIFICATIONS AND SITE ROUTING DETAILS FOR MORE INFORMATION.
7. ALL EQUIPMENT SHOWN IS AT FINISHED GRADE OR BELOW GRADE.



1 SITE UNDERGROUND POWER PLAN
3/64" = 1'-0"

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	ISSUES	
2	06.30.2022	ISSUE FOR PERMIT
3	10.27.2022	ADDENDUM #1 COMPOSITE
4	03.30.2023	COORDINATION SET
5	03.30.2023	ISSUE FOR CONSTRUCTION
6	02.06.2024	PHASE 2 & 3 / TFO
8	04.26.2024	BULLETIN #5

	REVISIONS	
5	02.16.2023	BULLETIN #1
6	03.30.2023	COORDINATION SET
7	05.22.2023	BULLETIN #2
8	07.11.2023	BULLETIN #3
9	02.06.2024	PHASE 2 & 3 / TFO
14	08.21.2024	BULLETIN #8

Contact: DAN HERRMANN Phone: 414.617.0382
Engineer: Addam Friedl, P.E.
License #: 049063 Exp. Date: 05.31.2025

COMMONWEALTH OF VIRGINIA ADDAM A. FRIEDL Lic. No. 049063 PROFESSIONAL ENGINEER

08.21.2024

COUNTY SEAL

NVA13

STACK INFRASTRUCTURE

9101 FREEDOM CENTER BLVD
MANASSAS, VA 20110

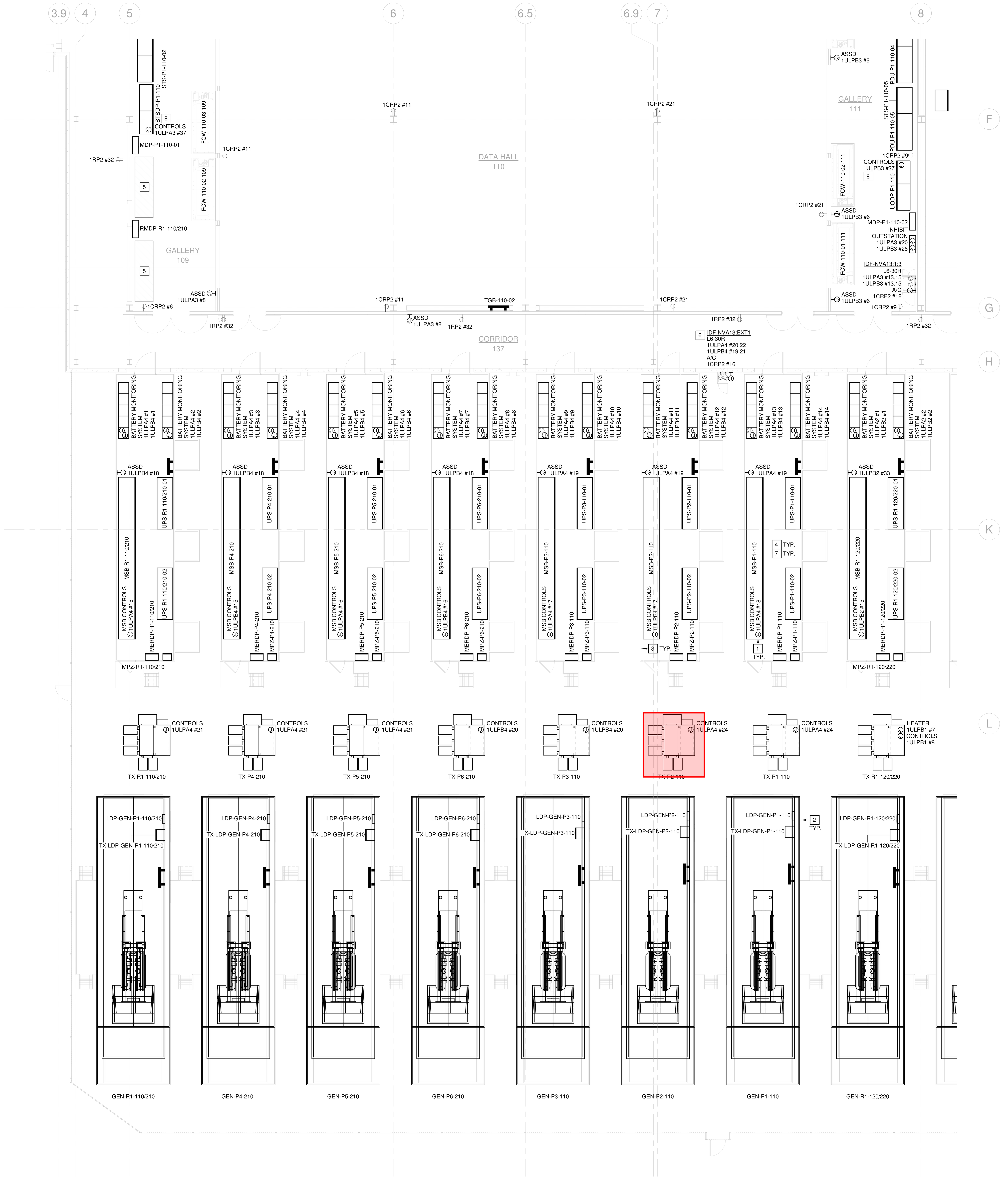
KEYPLAN

A B D F
C E G

SITE UNDERGROUND POWER PLAN

JOB 22.123
DATE 06.30.2022
SHEET

E00-03



1 POWER PLAN - LEVEL ONE - SEGMENT C
1/8" = 1'-0"

8/20/2024 4:50:31 PM E02-01C POWER PLAN - LEVEL ONE - SEGMENT C

GENERAL NOTES

1. REFER TO PROJECT GENERAL NOTES ON E00-01 FOR ADDITIONAL INFORMATION.
2. REFER TO ONE LINE AND PANEL SCHEDULE DRAWINGS FOR MECHANICAL EQUIPMENT INFORMATION.
3. INSTALL FLEXIBLE CONNECTIONS IN RUNS OF RACEWAYS, CABLES, WIREWAYS, CABLE TRAYS AND BUSWAYS WHERE THEY CROSS EXPANSION JOINTS, SEISMIC JOINTS OR ARE SUBJECT TO STRUCTURAL DEFLECTIONS OR INTERSTORY BUILDING DRIFT. FLEXIBLE CONNECTIONS SHALL BE INSTALLED WHERE ADJACENT SECTIONS OF RACEWAYS ARE SUPPORTED BY DIFFERENT STRUCTURAL ELEMENTS AND WHERE THEY TERMINATE TO EQUIPMENT THAT IS ANCHORED OR SUPPORTED TO A DIFFERENT STRUCTURAL ELEMENT. CONTRACTOR SHALL COORDINATE WITH STRUCTURAL AND/OR ARCHITECTURAL DRAWINGS IN ORDER TO PROVIDE THE REQUIRED FLEXIBLE CONNECTION.
4. VERIFY ALL EQUIPMENT LOCATIONS AND NAMEPLATE RATINGS FOR ALL OTHER TRADES EQUIPMENT PRIOR TO START OF ANY INSTALLATION.
5. WHEN FEASIBLE, ALL DRY TYPE TRANSFORMERS RATED 112.5KVA AND SMALLER SHALL BE HUNG FROM STRUCTURE ABOVE UNLESS OTHERWISE NOTED.
6. ALL WIRING WITHIN MECHANICAL GALLERIES IS TO BE DERATED FOR 95 DEGREES FAHRENHEIT INCLUDING BRANCH CIRCUITS, NOT SHOWN FOR CLARITY.

KEYED NOTES

1 ADDITIONAL REDUNDANT CONTROL CIRCUIT IS INTERNALLY DERIVED FROM MSB BY MSB MANUFACTURER.

2 GENERATOR ENCLOSURE MANUFACTURER TO PROVIDE EXTERIOR EMERGENCY STOP BUTTON WITH WEATHERPROOF COVER.

3 PROVIDE FULL SIZE ONE LINE DIAGRAM MOUNTED UNDER PLEXIGLASS COVER.

4 MER EQUIPMENT LOCATIONS ARE SHOWN TO ILLUSTRATE SCOPE OF WORK ONLY AND IS NOT LIMITED TO EQUIPMENT SHOWN. MER MANUFACTURER SHALL COORDINATE FINAL EQUIPMENT LAYOUT WITH CONTRACTOR PRIOR TO FABRICATION.

5 MAINTAIN CLEAR SPACE FROM ALL TRADES FOR FUTURE EQUIPMENT.

6 PROVIDE A GFCI CIRCUIT BREAKER AT THE BRANCH PANEL FOR EACH CIRCUIT.

7 MODULAR ELECTRICAL ROOM (MER) TO BE LISTED AND LABELED IN ACCORDANCE WITH NEC 110.3(A)(B)(C) BY MER MANUFACTURER. LISTING AND LABELING SHALL BE PRESENT FOR ELECTRICAL INSPECTION. MER IS NTA INSPECTED AND BEARS REGISTRATION AS AN INDUSTRIALIZED BUILDING UNIT WITH THE COMMONWEALTH OF VIRGINIA DEPARTMENT OF HOUSING AND COMMUNITY DEVELOPMENT.

8 ADDITIONAL REDUNDANT CONTROL CIRCUIT IS INTERNALLY DERIVED FROM SWITCHBOARD BY SWITCHBOARD MANUFACTURER.

MER INTERCONNECTION NOTES

EC TO PROVIDE ALL CONDUIT IN THIS PHASE AS REQUIRED FOR FUTURE CONNECTION FROM BUILDING/MER'S TO GENS/TRANSFORMERS/MVS'S.

EC TO COORDINATE WITH MER MANUFACTURER FOR ALL REQUIRED CONDUITS AND CIRCUITS ENTERING/LEAVING THE MER, INCLUDING, BUT NOT EXCLUSIVELY:

- **GEN AND MV TRANSFORMER OUTPUT FEEDERS:** UNDERGROUND THROUGH THE YARD FROM SOURCE TO MER.
- **CRITICAL OUTPUT FEEDERS:** BELOW GROUND UNDER BUILDING SLAB TO DATA HALL GALLERIES.
- **ACC OUTPUT FEEDERS:** ABOVE GROUND STUBBED THROUGH MER EAST WALL, UP THE BUILDING NORTH EXTERIOR WITHIN ARCHITECTURAL CHASE. REFER TO ARCHITECTURAL DRAWINGS.
- **GENERAL (PDP) AND DATA HALL CONDITIONING (MDP/RMDP) FEEDERS:** BELOW GROUND UNDER BUILDING SLAB FROM MSB TO DATA HALL GALLERIES AND ELEC ROOMS.
- **GENERATOR COMMS AND CONTROLS:** UNDERGROUND FROM MSB TO GENERATOR.
- **GENERATOR START/STOP SIGNAL:** UNDERGROUND FROM MSB TO GENERATOR.
- **GENERATOR EPMS/BMS:** UNDERGROUND FROM NEAREST IDF WITHIN BUILDING TO GENERATOR.
- **GENERATOR SHORE POWER:** UNDERGROUND FROM MSB TO GENERATOR
- **MV TRANSFORMER EPMS:** UNDERGROUND FROM NEAREST IDF WITHIN BUILDING TO MV TRANSFORMER.
- **MV TRANSFORMER AUXILIARY CIRCUITS:** UNDERGROUND FROM BUILDING INTERIOR TO MV TRANSFORMER.
- **PATCH PANEL:** ABOVE GROUND THROUGH BUILDING TO BUILDING MONITORING SYSTEM.
- **FIRE ALARM:** ABOVE GROUND THROUGH BUILDING TO BUILDING FIRE ALARM CONTROL PANEL.
- **SECURITY CAMERA:** ABOVE GROUND THROUGH BUILDING AS PART OF PATCH PANEL CONNECTION.
- **SECURITY DOOR:** DEDICATED ABOVE GROUND THROUGH BUILDING.
- **MER MISC 120V CIRCUITS:** ABOVE GROUND THROUGH BUILDING TO MER EQUIPMENT.

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	ISSUES	
1	05.13.2022	DESIGN DEVELOPMENT
2	06.30.2022	ISSUE FOR PERMIT
3	10.27.2022	ADDENDUM #1 COMPOSITE
4	03.30.2023	COORDINATION SET
5	03.30.2023	ISSUE FOR CONSTRUCTION
6	02.06.2024	PHASE 2 & 3 / TFO
7	03.27.2024	BULLETIN #4
8	04.26.2024	BULLETIN #5

	REVISIONS	
5	02.16.2023	BULLETIN #1
6	03.30.2023	COORDINATION SET
9	02.06.2024	PHASE 2 & 3 / TFO
10	03.27.2024	BULLETIN #4
11	04.26.2024	BULLETIN #5
13	06.14.2024	BULLETIN #7

Contact: DAN HERRMANN Phone: 414.617.0382
Engineer: Addam Friedl, P.E.
License #: 049063 Exp. Date: 05.31.2025

COMMONWEALTH OF VIRGINIA
ADDAM A. FRIEDL
Lic. No. 049063
PROFESSIONAL ENGINEER

08.21.2024

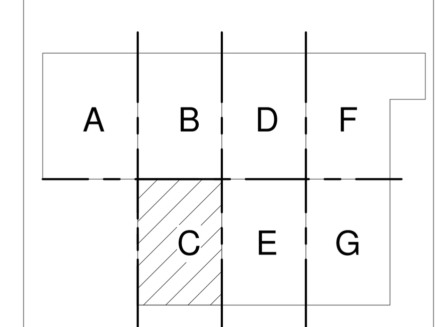
COUNTY SEAL

NVA13

STACK INFRASTRUCTURE

9101 FREEDOM CENTER BLVD
MANASSAS, VA 20110

KEYPLAN



POWER PLAN - LEVEL ONE - SEGMENT C

JOB 22.123
DATE 06.30.2022
SHEET

E02-01C

3 – Lift Worksheet(s)

Lift Worksheet

Title: Transformer Lift Plan
Date: 11/7/2024
Project: Stack NVA05A - Transformer (235T)
Job Number:
Description: Hoist & Set Transformer
Jobsite Address: 9101 Freedom Center Drive Manassas, VA
Customer: Rosendin Electric
P.O./ Contract#:
Lift Plan Drawing and Load Placement Drawing attached? Yes
Notes:

Crane Information

Manufacturer: Liebherr
Model: LTM 1200-5.1
Serial #: See Attached
Crane Rating: 235 t
Crane Inspection Date: Seee Attached
Notes:

Lift Information

Crane Radius: 95 ft
Crane Capacty at Radius: 24,200 lbs
Capacity at Pick Point: 24,200 lbs
Capacity at Set Point: 24,200 lbs
Notes:

Crane Configuration

Crane Carrier: 100% Outriggers (29' x 27')
Counterweight: 70,600 lbs
Chart Capacity: 24,200 lbs
Main Boom Length: 117' Telescopic Boom (T)
Boom Sections: 0-0-46-46-46-92
Parts of Line: 1
Line Size: 0.8"
Capacity of Line @ Parts: 23,600 lbs
Radius: 95 ft
Boom Angle: 31.3°
Tip Height: 71.3 ft
Jib Used? No
Jib: N/A
Jib Offset: N/A
Jib Angle from Ground: N/A
Outrigger Load: 81,264 lbs at 270° Swing Angle

Setup Information

Crane Setup: 360°
Setup Distance: 95'
Mat Used? Yes
Mat Dimensions: 8' L x 5.5' W x 8.75" H
Ground Bearing Pressure below Mat: 1,847 psf
Notes:

Load Configuration

Gross Load Weight: 17,717.5 lbs
Description: Transformer
Dimensions: 8 ft x 8 ft, Height: 8 ft
Load Weight: 16,500 lbs
Rigging Weight: 29.5 lbs
Hook Weight: N/A
Block Weight: 1,102 lbs
Load Line Weight: 86 lbs
Hook Height: 23.8 ft
Sling Length: 12'
Sling Angle: 62.4°
Sling Equipment #: ENR3
Sling Type: Endless Round Slings - Synthetic

Spreader Bar #: N/A
Spreader Bar Capacity: N/A
Hook Block: 23.1 Kips
Shackle Type: Crosby G-209
Shackle Qty: 4
Shackle Capacity: 3.25 Ton EA
Additional Rigging: N/A
Additional Rigging Capacity: N/A
% of Chart Capacity: 73%
Chart Capacity Deduction: N/A
Deduct Capacity: N/A
Notes:

Not for construction use. For pre-planning only.

Title: Transformer Lift Plan
Date: TBD
Project: Stack NVA05A - Transformer (235T)
Job Number:

Notes:

Notes:

Pre-Lift Checklist

Crane Operator: Name:

Signalperson Assigned: Name:

Communication Method:

Crane Inspected by Operator?	Yes No
Rigging Inspected?	Yes No
All Permits Obtained?	Yes No
Are weather conditions OK?	Yes No
Wind OK?	Yes No
Are there Power Lines?	Yes No
Is Operators Certification Card current?	Yes No
Is area OK for entry and exit of jobsite?	Yes No
Has a pre-lift meeting between operator, signalperson, supervisor, and any and all other persons occured?	Yes No

Other Considerations:

Signatures

Engineer:	Name:	Signature:	Date:
Supervisor:	Name:	Signature:	Date:
Operator:	Name:	Signature:	Date:
Client:	Name:	Signature:	Date:

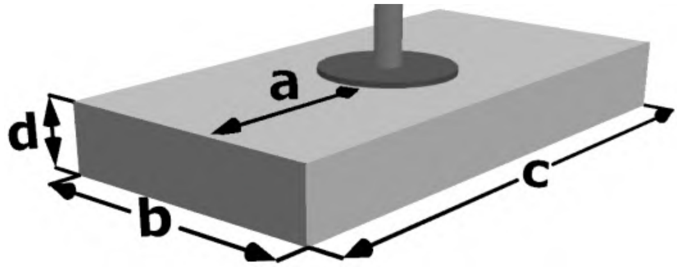
Not for construction use. For pre-planning only.

4 – Crane Mat Specifications

Ground Bearing Pressure Below Crane Mats

Job Information

Project	Stack NVA05A - Transformer (235T)
Customer	Rosendin Electric
Description	Hoist & Set Transformer
Drawn By	Hugo Moreno
Date	11/7/2024



Crane Information

Liebherr LTM 1200-5.1	
117' Telescopic Boom (T)	
100% Outriggers (29' x 27')	
Cwt: 70,600 lbs	
Outrigger Load 'P'	81,264.19 lbs
Outrigger Pad Length	4 ft
Outrigger Pad Width	3 ft

Mat Information

Mat Material	Steel
Mat Thickness	8.75 in
Mat Length	8 ft
Mat Width	5.5 ft
Mats in Width	1
Effective Mat Length 'c'	8 ft
Effective Mat Width 'b'	5.5 ft
Moment Arm 'a'	2.5 ft
Matting Layers 'n'	1
Total Mat Thickness 'd'	8.75 in

Calculations

Ground Bearing Pressure below mat...

$q = P / (b \cdot c)$ **1,847 psf**

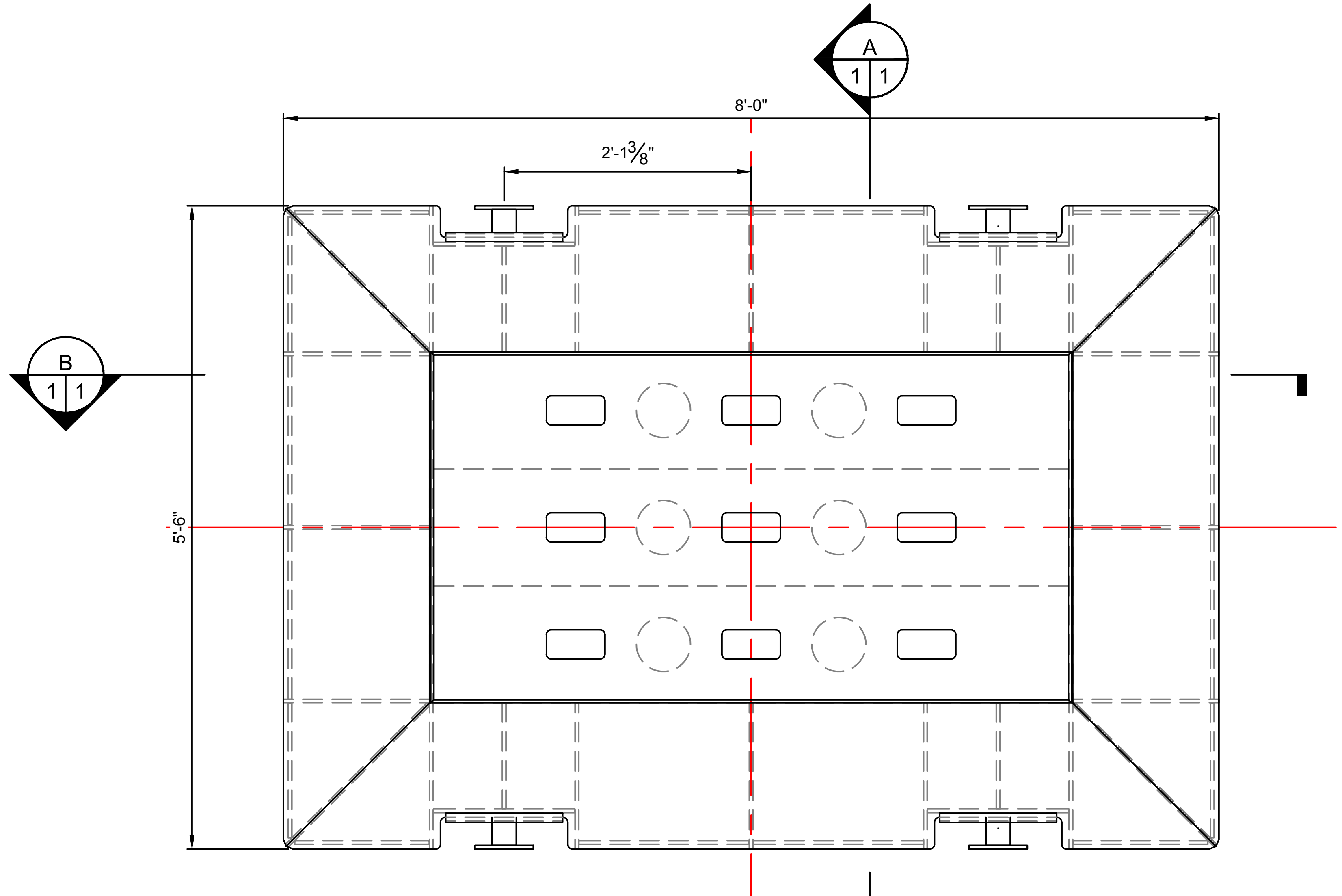
Bending Stress on Mat...

$f = 3 \cdot q \cdot a^2 \cdot n / d^2$ 452.3 psi **OK**

27,000 psi maximum allowed

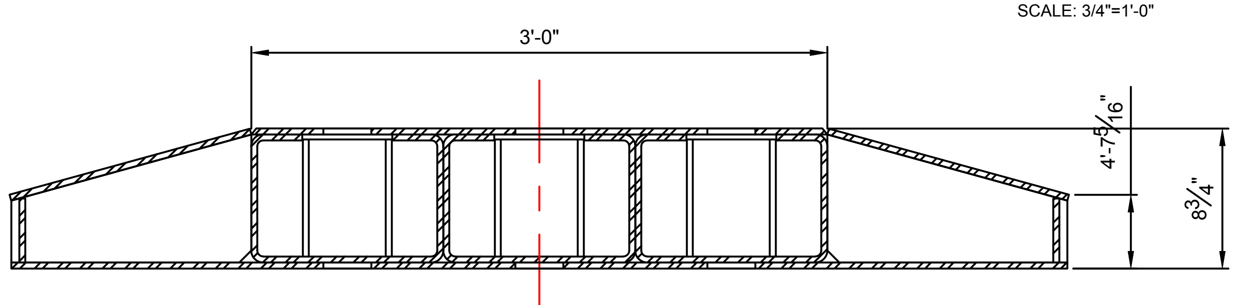
Notes

Not for construction use. For pre-planning only.



PLAN VIEW
SCALE: 3/4"=1'-0"

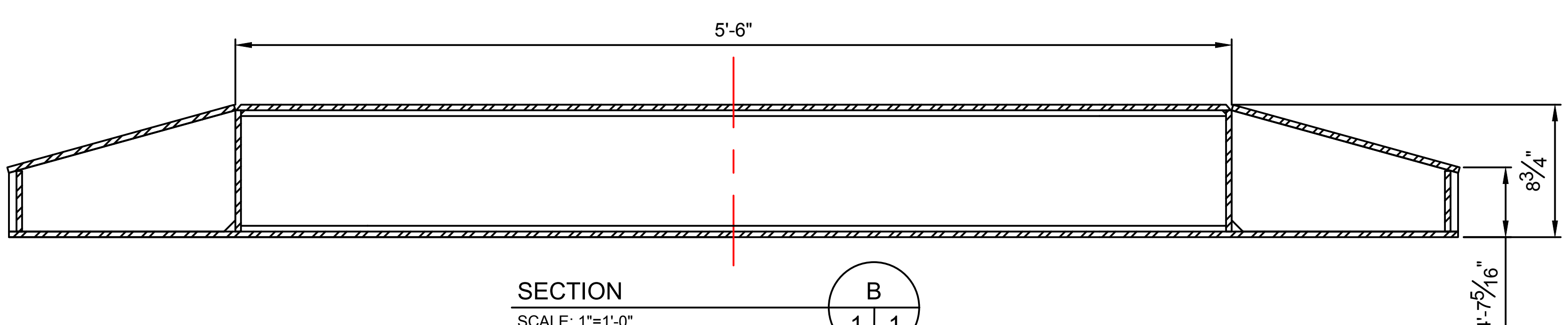
MAT SPECIFICATIONS:
APPROXIMATE WEIGHT: 2,600 LB.
CAPACITY: 235,000



SECTION A
SCALE: 1"=1'-0"

NOTES:

1. MAT CAPACITY IS BASED ON THE OUTRIGGER FLOAT BEING CENTERED ON THE MAT, WITH THE BOTTOM SURFACE OF THE MAT FULLY SUPPORTED. UNDER THESE CONDITIONS THE MAT IS CAPABLE OF DISTRIBUTING THE APPLIED LOAD EVENLY OVER ITS ENTIRE 5'-6"x8'-0" FOOTPRINT.



SECTION B
SCALE: 1"=1'-0"

REV	DESCRIPTION	DRAWN BY	DRAWN DATE	APPROVED BY	APPROVED DATE
0	FOR INFO	BPT	8/28/18		

W.O. GRUBB CRANE RENTAL

235K STEEL OUTRIGGER MAT - M-235-01
SPECIFICATION DRAWING
W.O. GRUBB FLEET EQUIPMENT
RICHMOND, VA

SCALE (U.N.O.):	SIZE:	JOB NO.	DRAWING NO.	SHEET NO.	REV. NO.
AS NOTED	B		M-235-01-SPEC	1 OF 1	0

5 – Crane Information & Inspection

Mobile Crane
Grue mobile

LTM 1200-5.1

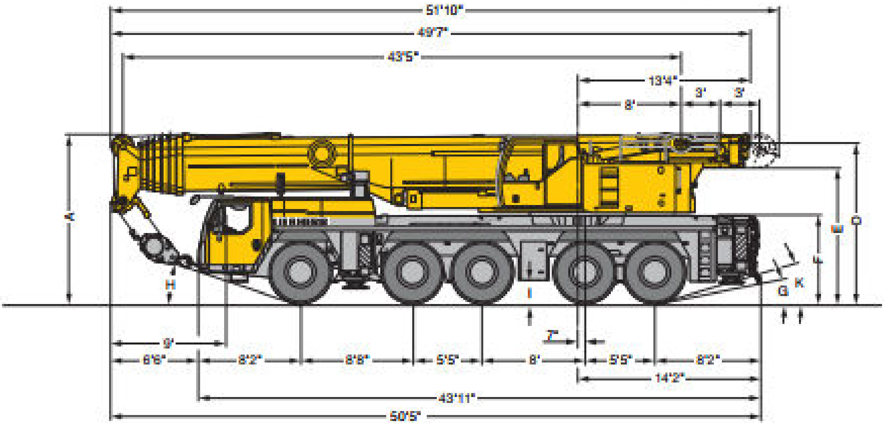
Technical Data
Caractéristiques techniques

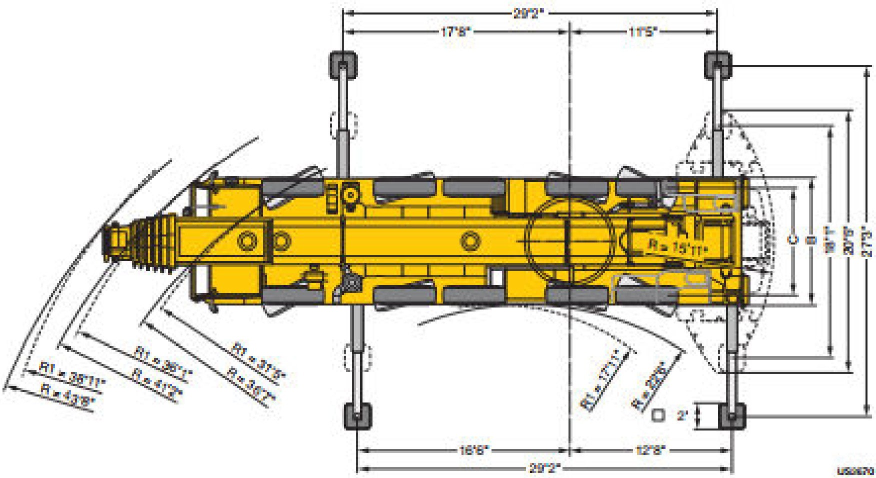
LICCON2

LIEBHERR

Dimensions
Encombrement

LICCON2





R₁ = All-wheel steering - Direction toutes roues

	Dimensions / Encombrement										
	A	A 0'6"*	B	C	D	E	F	G	H	I	K
445/95 R 25 (16.00 R 25)	13'1"	12'8"	9'10"	8'4"	12'4"	10'6"	6'9"	11"	17"	1'5"	16"
525/80 R 25 (20.5 R 25)	13'1"	12'8"	10'2"	8'5"	12'4"	10'6"	6'9"	11"	17"	1'5"	16"

* lowered - abaissé

Weights
Poids

LICCON2

Axle Essieu	1	2	3	4	5	Total weight lbs Poids total lbs
lbs	26400	26400	26400	26400	26400	132000

Load kips Forces de levage kips	No. of sheaves Poulies	No. of lines Brins	Weight lbs Poids lbs
332.9	9	16	4409
315.3	7	15	3306
238	5	11	2866
156.5	3	7	2292
69	1	3	1874
23.1	–	1	1102

Working speeds
Vitesses

	mph min.	mph max.	%
445/95 R 25 (16.00 R 25) 525/80 R 25 (20.5 R 25)	0.3	50	56 %

	12 / R2
	4 / R2

Drive Mécanismes	infinitely variable en continu		Rope diameter / Rope length Diamètre du câble / Longueur du câble	Max. single line pull Effort au brin maxi.
1	0 – 427	ft/min single line ft/min au brin simple	0.8" / 920 ft	23600 lbs
2	0 – 427	ft/min single line ft/min au brin simple	0.8" / 920 ft	23600 lbs
360°	0 – 1.3	rpm		
	approx. 65 seconds to reach 82° boom angle env. 65 s jusqu'à 82°			
	approx. 638 seconds for boom extension from 43 ft – 236 ft env. 638 s pour passer de 43 ft – 236 ft			

Load Chart

Project	Stack NVA05A - Transformer (235T)
Customer	Rosendin Electric
Description	Hoist & Set Transformer

Liebherr LTM 1200-5.1

Boom:	Telescopic Boom (T)
Jib:	-
Base:	100% Outriggers (29' x 27')
Counterweight:	70,600 lbs
Range:	360°
Capacity:	85%
Chart ID:	5411

Boom Section Percentages	Boom Length (ft)	Boom Angle	Jib Length (ft)	Jib Offset	Tip Height (ft)	Lift Radius (ft)	Capacity (lbs)	Note
46-46-46-46-46-0	117	80.6°	-	-	127.1	16	170,000	CODE 0014
46-46-46-46-46-0	117	79.6°	-	-	126.7	18	169,000	CODE 0014
46-46-46-46-46-0	117	78.6°	-	-	126.2	20	167,000	CODE 0014
46-46-46-46-46-0	117	76.6°	-	-	125.2	24	149,000	CODE 0014
46-46-46-46-46-0	117	74.6°	-	-	124	28	122,000	CODE 0014
46-46-46-46-46-0	117	72.5°	-	-	122.7	32	102,000	CODE 0014
46-46-46-46-46-0	117	70.4°	-	-	121.2	36	87,000	CODE 0014
46-46-46-46-46-0	117	68.3°	-	-	119.6	40	75,500	CODE 0014
46-46-46-46-46-0	117	64.5°	-	-	116.2	47	60,500	CODE 0014
46-46-46-46-46-0	117	60.6°	-	-	112.3	54	49,100	CODE 0014
46-46-46-46-46-0	117	56.4°	-	-	107.6	61	40,600	CODE 0014
46-46-46-46-46-0	117	52.1°	-	-	102.2	68	34,000	CODE 0014
46-46-46-46-46-0	117	47.5°	-	-	95.9	75	28,700	CODE 0014
46-46-46-46-46-0	117	42.5°	-	-	88.4	82	24,500	CODE 0014
46-46-46-46-46-0	117	36.9°	-	-	79.3	89	20,800	CODE 0014
46-46-46-46-46-0	117	30.3°	-	-	68	96	17,500	CODE 0014
46-46-46-46-46-0	117	22°	-	-	52.4	103	14,600	CODE 0014
92-46-46-46-0-0	117	80.6°	-	-	127.1	16	145,000	CODE 0014
92-46-46-46-0-0	117	79.6°	-	-	126.7	18	141,000	CODE 0014
92-46-46-46-0-0	117	78.6°	-	-	126.2	20	137,000	CODE 0014
92-46-46-46-0-0	117	76.6°	-	-	125.2	24	125,000	CODE 0014
92-46-46-46-0-0	117	74.6°	-	-	124	28	113,000	CODE 0014
92-46-46-46-0-0	117	72.5°	-	-	122.7	32	98,000	CODE 0014
92-46-46-46-0-0	117	70.4°	-	-	121.2	36	83,000	CODE 0014
92-46-46-46-0-0	117	68.3°	-	-	119.6	40	71,500	CODE 0014
92-46-46-46-0-0	117	64.5°	-	-	116.2	47	56,400	CODE 0014
92-46-46-46-0-0	117	60.6°	-	-	112.3	54	45,200	CODE 0014
92-46-46-46-0-0	117	56.4°	-	-	107.6	61	36,800	CODE 0014
92-46-46-46-0-0	117	52.1°	-	-	102.2	68	30,300	CODE 0014
92-46-46-46-0-0	117	47.5°	-	-	95.9	75	25,100	CODE 0014
92-46-46-46-0-0	117	42.5°	-	-	88.4	82	20,900	CODE 0014
92-46-46-46-0-0	117	36.9°	-	-	79.3	89	17,300	CODE 0014
92-46-46-46-0-0	117	30.3°	-	-	68	96	13,900	CODE 0014
92-46-46-46-0-0	117	22°	-	-	52.4	103	11,100	CODE 0014

Boom Section Percentages	Boom Length (ft)	Boom Angle	Jib Length (ft)	Jib Offset	Tip Height (ft)	Lift Radius (ft)	Capacity (lbs)	Note
0-46-46-46-46-46	117	80.6°	-	-	127.1	16	143,000	CODE 0014
0-46-46-46-46-46	117	79.6°	-	-	126.7	18	140,000	CODE 0014
0-46-46-46-46-46	117	78.6°	-	-	126.2	20	136,000	CODE 0014
0-46-46-46-46-46	117	76.6°	-	-	125.2	24	128,000	CODE 0014
0-46-46-46-46-46	117	74.6°	-	-	124	28	121,000	CODE 0014
0-46-46-46-46-46	117	72.5°	-	-	122.7	32	107,000	CODE 0014
0-46-46-46-46-46	117	70.4°	-	-	121.2	36	92,000	CODE 0014
0-46-46-46-46-46	117	68.3°	-	-	119.6	40	80,500	CODE 0014
0-46-46-46-46-46	117	64.5°	-	-	116.2	47	65,000	CODE 0014
0-46-46-46-46-46	117	60.6°	-	-	112.3	54	53,500	CODE 0014
0-46-46-46-46-46	117	56.4°	-	-	107.6	61	44,900	CODE 0014
0-46-46-46-46-46	117	52.1°	-	-	102.2	68	38,200	CODE 0014
0-46-46-46-46-46	117	47.5°	-	-	95.9	75	32,900	CODE 0014
0-46-46-46-46-46	117	42.5°	-	-	88.4	82	28,500	CODE 0014
0-46-46-46-46-46	117	36.9°	-	-	79.3	89	24,700	CODE 0014
0-46-46-46-46-46	117	30.3°	-	-	68	96	21,300	CODE 0014
0-46-46-46-46-46	117	22°	-	-	52.4	103	18,400	CODE 0014
0-0-92-46-46-46	117	80.6°	-	-	127.1	16	107,000	CODE 0014
0-0-92-46-46-46	117	79.6°	-	-	126.7	18	103,000	CODE 0014
0-0-92-46-46-46	117	78.6°	-	-	126.2	20	97,000	CODE 0014
0-0-92-46-46-46	117	76.6°	-	-	125.2	24	87,500	CODE 0014
0-0-92-46-46-46	117	74.6°	-	-	124	28	79,500	CODE 0014
0-0-92-46-46-46	117	72.5°	-	-	122.7	32	72,500	CODE 0014
0-0-92-46-46-46	117	70.4°	-	-	121.2	36	66,500	CODE 0014
0-0-92-46-46-46	117	68.3°	-	-	119.6	40	61,500	CODE 0014
0-0-92-46-46-46	117	64.5°	-	-	116.2	47	53,400	CODE 0014
0-0-92-46-46-46	117	60.6°	-	-	112.3	54	47,500	CODE 0014
0-0-92-46-46-46	117	56.4°	-	-	107.6	61	42,200	CODE 0014
0-0-92-46-46-46	117	52.1°	-	-	102.2	68	38,000	CODE 0014
0-0-92-46-46-46	117	47.5°	-	-	95.9	75	33,400	CODE 0014
0-0-92-46-46-46	117	42.5°	-	-	88.4	82	29,100	CODE 0014
0-0-92-46-46-46	117	36.9°	-	-	79.3	89	25,200	CODE 0014
0-0-92-46-46-46	117	30.3°	-	-	68	96	21,700	CODE 0014
0-0-92-46-46-46	117	22°	-	-	52.4	103	18,800	CODE 0014
0-0-46-46-46-92	117	80.6°	-	-	127.1	16	92,000	CODE 0014
0-0-46-46-46-92	117	79.6°	-	-	126.7	18	89,000	CODE 0014
0-0-46-46-46-92	117	78.6°	-	-	126.2	20	85,000	CODE 0014
0-0-46-46-46-92	117	76.6°	-	-	125.2	24	77,500	CODE 0014
0-0-46-46-46-92	117	74.6°	-	-	124	28	71,500	CODE 0014
0-0-46-46-46-92	117	72.5°	-	-	122.7	32	66,000	CODE 0014
0-0-46-46-46-92	117	70.4°	-	-	121.2	36	61,500	CODE 0014
0-0-46-46-46-92	117	68.3°	-	-	119.6	40	57,800	CODE 0014
0-0-46-46-46-92	117	64.5°	-	-	116.2	47	51,500	CODE 0014
0-0-46-46-46-92	117	60.6°	-	-	112.3	54	46,700	CODE 0014
0-0-46-46-46-92	117	56.4°	-	-	107.6	61	42,400	CODE 0014
0-0-46-46-46-92	117	52.1°	-	-	102.2	68	39,100	CODE 0014
0-0-46-46-46-92	117	47.5°	-	-	95.9	75	35,600	CODE 0014
0-0-46-46-46-92	117	42.5°	-	-	88.4	82	31,600	CODE 0014
0-0-46-46-46-92	117	36.9°	-	-	79.3	89	27,700	CODE 0014
0-0-46-46-46-92	**117**	**30.3°**	**-**	**-**	**68**	**96**	**24,200**	**CODE 0014**
0-0-46-46-46-92	117	22°	-	-	52.4	103	21,300	CODE 0014

Boom Section Percentages	Boom Length (ft)	Boom Angle	Jib Length (ft)	Jib Offset	Tip Height (ft)	Lift Radius (ft)	Capacity (lbs)	Note
0-0-0-46-92-92	117	80.6°	-	-	127.1	16	70,000	CODE 0014
0-0-0-46-92-92	117	79.6°	-	-	126.7	18	67,500	CODE 0014
0-0-0-46-92-92	117	78.6°	-	-	126.2	20	64,500	CODE 0014
0-0-0-46-92-92	117	76.6°	-	-	125.2	24	59,000	CODE 0014
0-0-0-46-92-92	117	74.6°	-	-	124	28	54,300	CODE 0014
0-0-0-46-92-92	117	72.5°	-	-	122.7	32	50,300	CODE 0014
0-0-0-46-92-92	117	70.4°	-	-	121.2	36	46,700	CODE 0014
0-0-0-46-92-92	117	68.3°	-	-	119.6	40	43,800	CODE 0014
0-0-0-46-92-92	117	64.5°	-	-	116.2	47	39,000	CODE 0014
0-0-0-46-92-92	117	60.6°	-	-	112.3	54	35,300	CODE 0014
0-0-0-46-92-92	117	56.4°	-	-	107.6	61	32,000	CODE 0014
0-0-0-46-92-92	117	52.1°	-	-	102.2	68	29,400	CODE 0014
0-0-0-46-92-92	117	47.5°	-	-	95.9	75	27,100	CODE 0014
0-0-0-46-92-92	117	42.5°	-	-	88.4	82	25,200	CODE 0014
0-0-0-46-92-92	117	36.9°	-	-	79.3	89	23,400	CODE 0014
0-0-0-46-92-92	117	30.3°	-	-	68	96	21,700	CODE 0014
0-0-0-46-92-92	117	22°	-	-	52.4	103	20,200	CODE 0014

This data is for reference use only. Operator must refer to in-cab charts to determine allowable lifting capacities.

CARSON CRANE, INC.

85 Main Street, Suite 5A, Reisterstown, MD 21136
(410) 526-7766 ● carsonCrane@msn.com

HYDRAULIC CRANE INSPECTION REPORT

CONSTRUCTION AND/OR INDUSTRY CERTIFICATION OF UNIT TEST AND/OR EXAMINATION OF CRANE, HOIST OR OTHER MATERIAL HANDLING DEVICE.

This documentation form is issued to record the survey herein noted, but does not substitute for any Federal or State certificates which may otherwise be required for the herein-listed equipment and which may be Recorded by another Carson Crane, Inc. certificate form for such Federal or State certification program.

Owner:	W.O. Grubb	Date:	4/4/2024
Contact:	Tom Boschi	Certificate Number:	240425
Address:	6331 South Hanover Road Elkridge, Maryland 21075	Inspector:	Tyler Clifton
		Expiration Date:	4/4/2025

Equipment: Crane Type: All Terrain

Location Of Equipment: Remains at Work Site: Changes Work Site: X
Describe Location: Equipment Yard

Manufacturer:	Liebherr	Model Number:	LTM1200-5.1
Serial Number:	093 374	Capacity:	235 Ton
Unit Number:	2350		

Service Status at Time of Survey: Lifting

Boom Status at Time of Survey: Length: 236’
Type: Hydraulic Telescopic

Test Loads Applied:	Radius	Proof Loads	Rated Loads
	N/A	N/A	N/A

Load Indicating or Limiting Device: Fitted Accuracy: Yes
Configuration at Time of Test: N/A
Application of Proof Loads: N/A
Basis for Assigned Load Ratings: Manufacturers Specifications
Remarks and/or Limitations: Survey at owner's request, and with his permission. ALL INFORMATION CORRECT AT TIME OF SURVEY.

Date of Last Inspection:
Previous Certification Number: N/A

I certify that on the 4th day of April 2024, the above-described device was examined as arranged by or with permission of the owner and that the equipment was noted to HAVE MET all applicable OSHA 1926.1400 regulations and ANSI B30.5 Standards.

Tyler Clifton

Date: 4/4/2024

Tyler Clifton

Neither Carson Crane or its representatives shall be liable in any manner for any personal injury or property damage of any kind arising from or connected to this survey. All satisfactory items are valid at time of inspection.

Carson Crane, Inc.
85 Main Street, Suite 5A
Reisterstown, Maryland 21136

Company	W.O. Grubb Crane Rental				
Make/ Model#	Liebherr LTM1200-5.1				
Serial#	093 374				
Unit#	2350	Capacity	235 Ton	Certification #	240425

General Appearance	S	U	NA	Comments
1. Glass	✓			
2. Sheet Metal	✓			
3. Guard Covers	✓			
4. Components	✓			
5. General Lubrication	✓			
6. Housekeeping	✓			
7. Safety/ Warning Decals & Labels	✓			
8. Paint Condition	✓			

Lower Cab	S	U	NA	Comments
9. Grab Rails & Steps	✓			
10. Instrument & Gauges	✓			
11. Electrical Switches & Functions	✓			
12. Horns/ Lights	✓			
13. Back-up Alarm	✓			
14. Operational Controls	✓			
15. Mirrors	✓			
16. Fire Extinguishers	✓			

Lower Engine	S	U	N/A	Comments
17. Leaks	✓			
18. Belts & Hoses	✓			
19. Air Induction System	✓			
20. Exhaust Systems	✓			
21. Radiator & Cooler System	✓			
22. Electrical Harness	✓			
23. Engine Mounts	✓			
24. Air Compressor	✓			
25. Batteries	✓			
26. General Condition	✓			
27. Miles			✓	
28. Hours			✓	

Drive Train	S	U	NA	Comments
29. Transmission Mounts	✓			
30. Transmission	✓			
31. Differential, Planetary Fluid			✓	
32. Axle Mounting & Suspension	✓			
33. Steering Linkage & Components	✓			
34. Tires & Lugs	✓			
35. Brakes	✓			

Carson Crane, Inc.
85 Main Street, Suite 5A
Reisterstown, Maryland 21136

Company	W.O. Grubb Crane Rental				
Make/ Model#	Liebherr LTM1200-5.1				
Serial#	093 374				
Unit#	2350	Capacity	235 Ton	Certification #	240425

Lower Frame	S	U	NA	Comments
35. Frame Members: Where Visible	✓			
36. Hoses & Connections	✓			
37. Electrical Wiring	✓			

Outriggers	S	U	NA	Comments
38. Boxes & Beams	✓			
39. Extensions & Cylinders	✓			
40. Hoses & Fittings	✓			
41. Hold Valves/ Position Locks	✓			

Upper Cab	S	U	NA	Comments
42. Grab Rails & Steps	✓			
43. Instruments & Gauges	✓			
44. Electrical Switches	✓			
45. Signal Horn	✓			
46. Operational Controls	✓			
47. Capacity Charts	✓			
48. Control Labels	✓			
49. Warning Decals/ Signal Chart	✓			
50. Mirrors	✓			
51. Air/ Hydraulic Leakage	✓			
52. Crane Level Indicator	✓			
53. Anti-Two Block Device	✓			
54. Boom Angle Indicator	✓			
55. Load Weight Device	✓			
56. Fire Extinguisher	✓			
57. Operators Manual	✓			

Upper Engine	S	U	N/A	Comments
58. Leaks	✓			
59. Belts & Hoses	✓			
60. Air Induction System	✓			
61. Exhaust System	✓			
62. Radiator & Cooler System	✓			
63. Electrical Harness	✓			
64. Batteries	✓			
65. Air Compressor/Governor	✓			
66. Hours			✓	

Carson Crane, Inc.
85 Main Street, Suite 5A
Reisterstown, Maryland 21136

Company	W.O. Grubb Crane Rental				
Make/ Model#	Liebherr LTM1200-5.1				
Serial#	093 374				
Unit#	2350	Capacity	235 Ton	Certification #	240425

Rotating Upper Structure	S	U	NA	Comments
67. Frame Condition: Where Visible	✓			
68. Turntable: VISUAL	✓			
59. Electrical Collector Ring:	✓			
70. Hydraulic/ Air/ Electrical Leaks:	✓			
71. Gear Box:	✓			
72. Winch Mounting:	✓			
73. Winch Motor, Brake, Valve, Line:	✓			
74. Attachment Mounts, Pins, Keepers:	✓			
75. Counterweight Mount:			✓	

Main Boom	S	U	N/A	Comments
76. Lift Cylinder(s)	✓			
77. Telescoping Cylinder(s)	✓			
78. Hydraulic Hoses/Tubing Fittings	✓			
79. Holding Device	✓			
80. Boom Sections Alignment	✓			
81. Wear Pads	✓			
82. Sheaves	✓			
83. Hoist Line Dead End	✓			
84. Wire Rope Retainer	✓			
85. Boom Hinge Pin	✓			
86. Boom Head Section	✓			
87. Structure	✓			

Jib Extension	S	U	NA	Comments
88. Positive Stop(s)			✓	
89. Sheave(s)			✓	
90. Wire Rope Retainer(s)			✓	
91. Structure			✓	

Carson Crane, Inc.
85 Main Street, Suite 5A
Reisterstown, Maryland 21136

Wire Rope, Block, and Hook Report

Date Installed	N/A
Certificate on Hand	No
Wire Rope Maintained	Satisfactory
Description	Non-rotating

Application of Wire Rope

Parts of Line	2	Broken Wires	No
Measured Diameter	23mm	Rope Damage	No
Excessive Wear	No	Sheaves & Drums	Satisfactory
Main Hoist	Satisfactory		

Main Block & Hook

Manufacturer	Liebherr
Model	32 Ton
Weight	840 kg
Description	
Tram Measurement	
NDT	

Auxiliary Block & Hook

Manufacturer	
Model	
Serial	
Description	
Tram Measurement	
NDT	

Hoisting From:	Boom/Jib Length	Load Radius	Boom Angle	Parts of Line	Rated Capacity	Test Weight	% of Rated Capacity
Boom	236’						
Jib							
No-Load Operational Load Test	Satisfactory	**Caution:** Operation of cranes by Inspectors shall be limited to those crane functions necessary to accomplish the inspection. Inspectors must have met all operator trainee qualification requirements outlined by ANSI/ASME. B30.5 i.e. passed physical and written examinations.					

Carson Crane, Inc.
85 Main Street, Suite 5A
Reisterstown, Maryland 21136

Deficiency Report

The following corrective action(s) (repairs, adjustments, replacement parts, etc) are to performed by a qualified person in accordance with all manufacturer's instructions, specifications, and requirements. OSHA requires that, *"any deficiency shall be repaired or defective parts replaced before continued use."*

Item#	Explanation
	No deficiencies noted at time of survey.

Carson Crane, Inc.
85 Main Street, Suite 5A
Reisterstown, Maryland 21136

Recommendation Report

The following recommendation(s) should be considered for corrective action. The advisability of which depends on the facts in each situation. Any corrective action(s) are to be performed by a qualified person in accordance with all the manufacturer's instructions, specifications, and requirements.

Item#	Explanation	Immediate Action Required?
	No recommendations at this time.	

6 – Crane Operator Certifications

6331 South Hanover Road
Elkridge, MD 21075

410-796-3661
Fax: 410-796-3666

Personnel Certifications: **Brandon Ashley**

CCO CERTIFIED
CCO ID: **444128954**

Scan QR code or go to www.verifycco.org to verify current certification(s).

BRANDON C. ASHLEY

Committed to Quality, Integrity, and Fairness since 1995

MARYLAND USA
Commercial Driver's License
CDL

A10559A03

Customer identifier: MD-10274658695
Family name: ASHLEY
Given names: BRANDON CHESTER
Address: 1522 BROWN RD, WESTMINSTER MD 21158-2037
Date of birth: 01/26/1988 | Sex: M | Height: 6'-01" | Weight: 220 | Date of exp: 01/26/2032
Restrictions: | Classifications: AM | Endorsements: NT | Date of issue: 12/10/2023
ORGAN DONOR

Form MCSA-5876

OMB No. 2126-0006 Expiration Date: 12/31/2024

U.S. Department of Transportation
Federal Motor Carrier Safety Administration

Medical Examiner's Certificate
(for Commercial Driver Medical Certification)

I certify that I have examined Last Name: Ashley First Name: Brandon in accordance with (please check only one):

- ⦿ the Federal Motor Carrier Safety Regulations (49 CFR 391.41-391.49) and, with knowledge of the driving duties, I find this person is qualified, and, if applicable, only when (check all that apply) OR
- ○ the Federal Motor Carrier Safety Regulations (49 CFR 391.41-391.49) with any applicable State variances (which will only be valid for intrastate operations), and, with knowledge of the driving duties, I find this person is qualified, and, if applicable, only when (check all that apply):

☐ Wearing corrective lenses ☐ Accompanied by a ______ waiver/exemption ☐ Driving within an exempt intracity zone (49 CFR 391.62) (Federal)
☐ Wearing hearing aid ☐ Accompanied by a Skill Performance Evaluation (SPE) Certificate ☐ Grandfathered from State requirements (State)

The information I have provided regarding this physical examination is true and complete. A complete Medical Examination Report Form, MCSA-5875, with any attachments, embodies my findings completely and correctly, and is on file in my office.

Medical Examiner's Certificate Expiration Date: 02/15/2026

Medical Examiner's Signature:
Medical Examiner's Telephone Number: (410) 882-0808
Date Certificate Signed: 02/15/2024
Medical Examiner's Name (please print or type): Joseph Kligman
⦿ MD ○ Physician Assistant ○ Advanced Practice Nurse
○ DO ○ Chiropractor ○ Other Practitioner (specify)
Medical Examiner's State License, Certificate, or Registration Number: D29601
Issuing State: Maryland
National Registry Number: 9456288338

Driver's Signature:
Driver's License Number: MD-10274658695
Issuing State/Province: Maryland
Driver's Address: Street Address: 1522 Brown Rd City: Westminister State/Province: MD Zip Code: 21158
CLP/CDL Applicant/Holder ⦿ Yes ○ No

This document contains sensitive information and is for official use only. Improper handling of this information could negatively affect individuals. Handle and secure this information appropriately to prevent inadvertent disclosure by keeping the documents under the control of authorized persons. Properly dispose of this document when no longer required to be maintained by regulatory requirements.

OSHA
Occupational Safety and Health Administration

14-006309849

This card acknowledges that the recipient has successfully completed:

10-hour Construction Safety and Health

This card issued to:
Brandon Ashley

Quinton Anderson — Trainer Name
4/11/2022 — Date of Issue

IUOE NTF
National HAZMAT Program

Brandon Ashley
completed the following training

Globally Harmonized System (GHS)
of Classifcation and Labeling of Chemicals awareness course

GHS includes the new hazards classification criteria, pictograms, label elements and safety data sheets in accordance with OSHA 29 CFR 1910.1200, Hazard Communication (HAZCOM) 2012

John Simons, Jr — Instructor
Completion Date: 10/9/13

IUOE NTF - National HAZMAT Program

6331 South Hanover Road
Elkridge, MD 21075

410-796-3661
Fax: 410-796-3666

Personnel Certifications: **Brandon Ashley**

Rigger Qualification
Crane Signaling Qualification

This Card Verifies

Brandon Ashley

Has Successfully met the requirements for **Qualified Rigger** and **Qualified Crane Signal Person** In Accordance with all applicable **ASME B30** Safety Standards for **Cranes and Hoists** and all applicable **OSHA Safety Regulations** 29 CFR 1926 for **Construction** as indicated on the reverse.

Employee ID# 1279 ***Expiration Date*** 01/26/2027

The employee named on this card:

Has successfully completed a Crane Signal Qualification Course In accordance with **OSHA 29CFR 1926.1919 - 1926.1922 Signals** and meets the requirements of **1926.1428 - Signal Person Qualifications**

Trained In: Hand - Voice - Audible Signals

and has successfully completed a Rigging Qualification Course in accordance with:

- **ASME B30.5** Mobile and Locomotive Cranes
- **ASME B30.26** Rigging Hardware
- **ASME B30.5** Tower Cranes
- **ASME B30.6** Derricks
- **ASME B30.9** Slings
- **ASME B30.23** Personnel Lifting Systems
- **OSHA 29 CFR 1926.251** Rigging Equipment for Material Handling
- **OSHA 29 CFR 1926.1431** Hoisting Personnel

and is qualified as a:
Rigger Level II

Quinton B. Anderson
Instructor

ForkLift Operator Safety Training

Issue Date: 01/26/2024

Expiration Date: 01/26/2027

Brandon Ashley

Has Successfully Completed the W.O. Grubb ForkLift Safety Training Qualification Course for Written and Practical Evaluation In Accordance With **OSHA** Regulation 29 CFR§1910.178 Powered Industrial Trucks

Instructor: Quinton B. Anderson

The named Individual on the face of this card has successfully completed practical evaluation in the following classes:

- ☐ Class I
- ☐ Class II
- ☐ Class III
- ✓ Class IV
- ✓ Class V
- ☐ Class VI
- ✓ Class VII

In accordance with OSHA Regulation 29 CFR §1910.178 Powered Industrial Trucks Additional site specific operator evaluation is required prior to operating a forklift in the work place.

Fall Protection Training Certification

This Card Certifies That:

Brandon Ashley
Employee Name

FALL PROTECTION TRAINED

Has been trained in accordance with OSHA 29 CFR 1926 Subpart M Fall Protection and OSHA 29 CFR Subpart CC Cranes and Derricks in Construction 1926.1423 Fall Protection.

Instructor: Quinton B. Anderson ***Employee ID#*** **1279**

CRANE OPERATOR QUALIFICATION TRAINING REPORT

ver 12-27-2016
revised 11/8/2018cdc – revised 3-26-2021

WO GRUBB CRANE RENTAL

Operator being evaluated: Brandon Ashley **Date:** 3/8/24

Crane make and model(s) if similar: Select from the list on the next page by placing a check mark in the box.

Crane Type: ☐ Lattice Boom Crawler ☐ Lattice Boom Truck ☐ Tele Crawler ☑ All Terrain Hydraulic crane ☑ Hydraulic Truck Crane ☐ Rough Terrain Crane ☑ Boom Truck

Certification Type: ☐ LBT ☐ LBC ☑ TSS ☑ TLL ☐ LBT ☑ BTF ☐ STC

(MSHA alternate 5000-23 for Part 46 and 48)
☑ **Record and Certification of Training**
☐ **Annual Refresher Training Certificate**
☑ **New Task Training Certificate**

This form complies with OSHA 29CFR1926.1427 operator Training, certification and evaluation/qualification

Branch: ☐ Richmond ☐ Alexandria ☑ Baltimore ☐ Portsmouth ☐ Fredericksburg ☐ LCD ☐ Greensboro ☐ Winchester ☐ Space Road ☐ Roanoke ☐ Wilmington ☐ Newport News

Manufactures Safe Operating Procedures	Date Comp.	Crane Configuration	Date Completed
☑ LMI System	3/8/24	☑ Main boom	3/8/24
☑ Out Rigger Controls	3/8/24	☑ Main boom and jib	3/8/24
☑ Assembly/Disassembly	3/8/24	☐ Main boom and luffing	
☑ Jib Erection/Stowing	3/8/24	☐ Main boom with luffing and y guy/mega wing/SSL	
☐ Luffing Assembly/Disassembly	3/8/24	☐ Main boom and y guy, mega wing, SSL or heavy lift	
☑ Counterweight Configuration	3/8/24	☐ Main boom and sled, ballast wagon or VPC MAX	
☑ Load Charts	3/8/24	☐ Main boom with luffing and sled, ballast wagon or VPC MAX	
☑ Operators Manual	3/8/24	☑ Variable Select outriggers	3/8/24
☑ Site Driving crane erected	3/8/24	☐ Special attachments (clam bucket, pile driving leads etc.....)	

Other Training	Date Comp.	Employee Signature
OSHA Sub-Part CC Crane Standards		
Rigging		
Crane Signals: Voice, Hand Audible, Radio		

Employee Signature Below Verifying That Training Was Received

Operator Printed Name and Signature: Brandon Ashley / Brandon Ashley

Trainer/Evaluator printed name: Quinton Anderson

Trainer/Evaluator Signature: Quinton B. Anderson **Date:** 3/8/2024

Operations Manager Signature: [signature] **Date:** 3/8/2024

Signature of Person Designated in the Training Plan as Responsible for Safety & Health Training at the Mine)
I CERTIFY THAT THE ABOVE TRAINING HAS BEEN COMPLETED

False Certification Is Punishable Under 110 (a) and (f) of the Federal Mine Safety and Health Act

Crane Type - Manufacturer - Model

All Terrain Cranes

	Manufacturer	Model	Capacity
	Liebherr	LTM-1060/2	73 ton
	Liebherr	LTM-1095-5.1	115 ton
	Grove	GMK-5120B	120 ton
✓	Grove	GMK-5165	165 ton
✓	Grove	GMK-5165/2	165 ton
	Demag	AC-140	170 ton
✓	Grove	GMK-5150L	175 ton
	Link-Belt	ATC-175	175 ton
	Demag	AC-160/2	200 ton
	Link-Belt	ATC-3210	210 ton
✓	Liebherr	LTM-1200/5.1	235 ton
	Liebherr	LTM-1250/1	300 ton
	Liebherr	LTM-1250/6.1	300 ton
	Grove	GMK-6350	350 ton
	Grove	GMK-6300L	350 ton
	Liebherr	LTM-1300/1	365 ton
	Demag	AC-350/6	400 ton
	Liebherr	LTM-1350/6.1	400 ton
	Grove	GMK-7550	550 ton
	Liebherr	LTM 1450-8.1	550 ton
	Liebherr	LTM-1500/8.1	600 ton
	Demag	AC-500/8	600 ton

Carry Deck Cranes

	Manufacturer	Model	Capacity
	Grove	AP-206	6 ton
	Grove	AP-308	8.5 ton
	Shuttlelift	330-ELB	8.5 to
	Terex	D-851	8.5 ton
	Broderson	IC-200-3G	15 ton
	Broderson	IC-250-3B	18 ton
	Broderson	IC-250-3D	18 ton
	Grove	GDC-25	25 ton
	Grove	YB-7725	25 ton
	Shuttlelift	CD-7725	25 ton

Boom Trucks

	Manufacturer	Model	Capacity
	Manitex	2264	22 ton
	Manitex	22101-S	22 ton
✓	Terex Stinger	4792	23.5 ton
	USTC	2500JBT	25 ton
	Manitex	26101C	26 ton
	Manitex	2892C	28 ton
	Manitex	30100C	30 ton
	Manitex	30102C	30 ton
	National	14110	33 ton
	Manitex	35124C	35 ton
✓	Manitex	40124C	40 ton
	National	NTC 55	55ton

Hydraulic Truck Cranes

	Manufacturer	Model	Capacity
✓	Grove	TMS-500E	40 ton
	Link-Belt	HTC-8640HL	40 ton
	Link-Belt	HTC8650	50 ton
✓	Grove	TMS-700E	60 ton
✓	Link-Belt	HTC5660 III	60 ton
	Terex	T560/1	60 ton
✓	Link-Belt	HTC-8675 II	75 ton
	Terex	T-775	75 ton
✓	Grove	TMS-800E	80 ton
✓	Grove	TMS-900E	90 ton
	Link-Belt	HTC-8690	90 ton
	Link-Belt	HTC-86100	100 ton
✓	Grove	TMS-9000E	110 ton
	Link-Belt	HTC-86110	110 ton
	Grove	TMS-9000-2	115 ton

Crawler Cranes

	Manufacturer	Model	Capacity
	Terex	HC80	80 ton
	Link-Belt	LS-218H	100 ton
	Link-Belt	218HSL	110 ton
	Terex/American	HC-110	110 ton
	Manitowoc	12000	120 ton
	Link-Belt	238HSL	150 ton
	Link-Belt	LS-238H	150 ton
	Terex	HC-165	165 ton
	Link-Belt	LS-248HSL	200 ton
	Link-Belt	LS-248HV	200 ton
	Link-Belt	LS-248H	200 ton
	Link-Belt	298HSL	250 ton
	Link-Belt	LS-278H	250 ton
	Manitowoc	999	300 ton
	Manitowoc	2250	275 ton
	Liebherr	LR-1300SX	300 ton
	Terex/Demag	CC-1500	330 ton
	Liebherr	LR-1350/1	380 ton
	Terex/Demag	CC-2200	385 ton
	Manitowoc	MLC-300	386 ton
	Manitowoc	16000	440 ton
	Liebherr	LR-1400/2	450 ton
	Liebherr	LR-1750/2	825 ton

Lattice Boom Truck Cranes

	Manufacturer	Model	Capacity
	Link-Belt	HC-148H	200 ton
	Link-Belt	HC—278H	300 ton

Rough Terrain Cranes

	Manufacturer	Model	Capacity
	Terex	CD-218	18 ton
	Grove	RT-528C	28 ton
	Grove	RT-530E	30 ton
	Grove	RT-535E	35 ton
	Terex	RT-335/1	35 ton
	Grove	RT-540E	40 ton
	Link-Belt	RTC-8040 II	40 ton
	Grove	RT-650E	50 ton
	Link-Belt	RTC-8050 II	50 ton
	Tadona	TR-500XL/4	50 ton
	Terex	RT-555/1	55 ton
	Grove	RT-760E	60 ton
	Tadona	TR-600XXL	60 ton
	Terex	RT-160	60 ton
	Link-Belt	RTC-8065	65 ton
	Link-Belt	RTC-8065 II	65 ton
	Terex	RT-670	70 ton
	Grove	RT-875E	75 ton
	Grove	RT-875BXL	75 ton
	Link-Belt	RTC-8075	75 ton
	Link-Belt	RTC-8080 II	80 ton
	Terex	RT-780	80 ton
	Grove	RT-890E	90 ton
	Link-Belt	RTC-8090 II	90 ton
	Link-Belt	RTC-80100 II	100 ton
	Link-Belt	RTC-80110 II	110 ton
	Link-Belt	100 RT	110 ton
	Grove	RT-9130E	130 ton
	Link-Belt	RTC-80130 II	130 ton

Telescopic Crawler Cranes

	Manufacturer	Model	Capacity
	Link-Belt	TCC-1150	75 ton
	Link-Belt	TCC-1100	110 ton

False Certification Is Punishable Under 110 (a) and (f) of the Federal Mine Safety and Health Act

7 – Rigger Certifications

A&A TRANSFER
An MEI Company

Employee Qualification Cards FIELD

5/3/2023

Christopher M. Abrams
Employee ID# 000185
Warehaouse

An MEI Company
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338

Joe Adams III
Employee ID# 000235
Warehaouse

Ryan J. Adler
Employee ID# 000186
Dir. of Project Management

Luis A. Alvarez
Employee ID# 000094
Rigger

Alexander Amaya Portillo
Emp ID #101976
Rigger

Fermin A. Amaya
Employee ID# 000056
Lead Man

Mike A. Amaya
Employee ID# 102065
Rigger Trainee

Joseph N. Ammah
Employee ID# 000108
Driver

Imber I Andrade
Employee ID# 102027
Driver

Kevin G. Andrade
Employee ID# 101983
Driver

Jhony M. Arellano
Employee ID# 000219
Driver

Omar R. Avendano
Employee ID# 000202
Driver/Rigger Trainee

Sergio R. Avendano
Employee ID# 000085
Lead Man

Jackson R. Ballard
Employee ID# 000084
Driver

Employee Qualification Cards FIELD


Bladimir A. Barrera
Employee ID# 001154
Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Emerson B. Barrera
Employee ID# 000169
Driver/Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Robert P. Block Jr.
Employee ID# 102005
Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Michael J. Bonafede III
Employee ID# 000198
Driver/Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Calvin D. Bota
Employee ID# 102069
Warehouse Ohio
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Spencer O. Bowers
Employee ID# 102019
Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Calvin M. Boyce Jr.
Employee ID# 000124
Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


James P. Brennskag
Employee ID# 000269
Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Herbert C. Brensinger IV
Employee ID# 101996
Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Everett L. Brewster
Employee ID# 000087
Warehouse Manager
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338

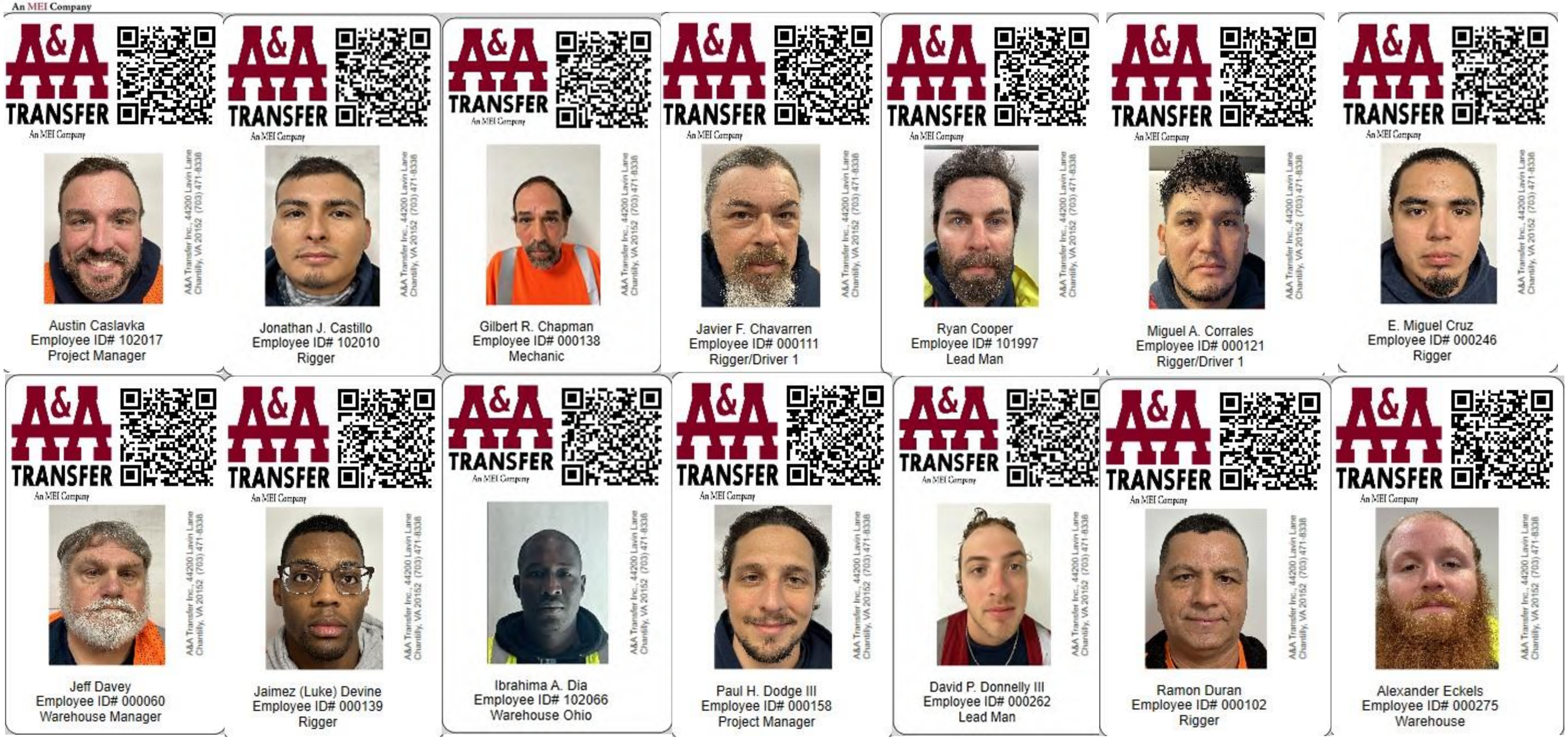

Cameron A. Brewton
Employee ID# 102023
Mechanic
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Edward E. Brock
Employee ID# 000261
Warehouse
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Luis A. Callejas
Employee ID# 000171
Driver/Rigger
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338


Thomas R. Carver Jr.
Employee ID# 101991
Lead Man
A&A Transfer Inc., 44200 Lavin Lane Chantilly, VA 20152 (703) 471-8338

Employee Qualification Cards FIELD



Employee Qualification Cards FIELD

John P. Ellerbe
Employee ID# 102067
Warehouse Ohio

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Victor M. Escobar-Torres
Employee ID# 000059
Warehouse

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Mynor Estrada
Employee ID# 000054
Lead Man

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Kevin Estrada-Arevalo
Employee ID# 102073
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Stephen B. Fannin
Employee ID# 000062
Rigger/Driver 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Joshua D. Flouhouse
Employee ID# 102001
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Edwin E. Funez
Employee ID# 102078
Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Noah Furr
Employee ID# 000148
Inventory Secialist

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Jorge Garcia de leon
Employee ID# 102038
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Jesse Garrison
Employee ID# 101973
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Brian M. George
Employee ID# 102033
Warehouse Manager Ohio

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Mario Gonzalez Medrano
Employee ID# 102029
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Employee Qualification Cards FIELD

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Cornelio Gonzalez Teo
Employee ID# 000240
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Frank Bible Graham
Employee ID# 102052
CDL Driver/Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Milter E. Guevara Montesflores
Employee ID# 000152
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Estuardo Guevara
Employee ID# 000070
Lead Man

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

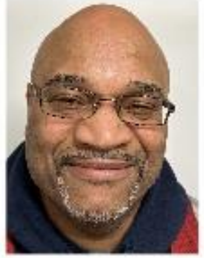
Jorge Guzman Hernandez
Employee ID# 102007
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Kyle C. Hanson
Employee ID# 000099
Project Manager

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Michael Hercules
Employee ID# 101974
Rigger



A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Charles W. Hogan Jr.
Employee ID# 000066
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Joshua E. Honore
Employee ID# 102079

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Akrem Idris
Employee ID# 102031
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Rolando S. Iriarte
Employee ID# 001152
Rigger/Driver

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

M. Feeroze Jaghory
Employee ID# 000190
Rigger

Employee Qualification Cards FIELD

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Luis O. Jovel-Majano
Employee ID# 102032
Rigger/Driver Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Matthew Gage Keiter
Employee ID# 000179
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Zachary Kennedy-Noce
Employee ID# 102015
Mechanic

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Brandon W. Kidwell
Employee ID# 000239
Driver

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Aaron Kinney
Employee ID# 000107
Project Manager

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Nathaniel L. Lamptey
Employee ID# 000093
Rigger/Driver 3

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Javier Latalladi
Employee ID# 102063
Warehouse

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Fidel R. Limon Cuellar
Employee ID# 000129
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Gilberto Limon Cuellar
Employee ID# 000260
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Matthew Lindsay
Employee ID# 102075
Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Allan Lopez Umana
Employee ID# 102040
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Dominic A Luis
Employee ID# 000178
Rigger

A&A TRANSFER
An MEI Company

Employee Qualification Cards FIELD

Joshua P. Luis
Employee ID# 000092
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Austin W. Lusk
Employee ID# 101980
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Aaron J. Maines
Employee ID# 000103
Lead Man

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Adelmo D. Marchorro Revolorio
Employee ID# 102054
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Jonathan L. Marks
Employee ID# 000276
Warehouse

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Anthony Marquez Diaz
Employee ID# 102021
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Carlos M. Martinez Aragon
Employee ID# 101998
Rigger/Driver 2

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

James D. McDonald
Employee ID# 000080
Project Manager

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Dustin C. McIntosh
Employee ID# 000075
Project Manager

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Kendall I. McKnight
Employee ID# 102080
Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Troy Mearkle
Employee ID# 000173
Warehouse/Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Nestor J. Merino
Employee ID# 102076
Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8338

Employee Qualification Cards FIELD

Employee Qualification Cards FIELD

Carlos Padilla-Quiroz
Employee ID# 000132
Driver

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Henry O. Palacios
Employee ID# 102071
Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

James K. Papay
Employee ID# 000231
Rigger 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Joseph Pauls
Employee ID# 101999
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Adam Piwetz
Employee ID# 102055
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Edward P. Ponzini
Employee ID# 000255
Warehouse

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Christian M. Powell
Employee ID# 102004
Asst. Ops Manager

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Scott A. Presti
Employee ID# 000177
Warehouse Manager

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Dylan Regan-Wiley
Employee ID# 102025
Rigger Trainee

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Kevin A. Recinos-Rivera
Employee ID# 000265
Rigger

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

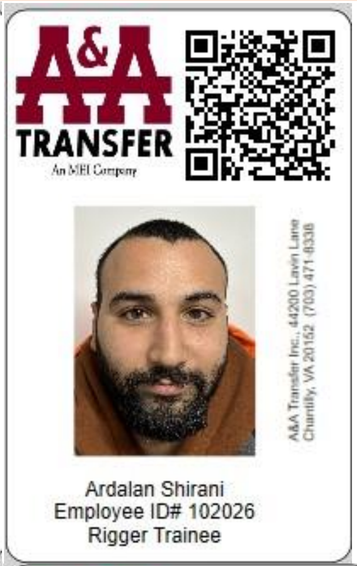
Jose V. Rivera Garcia
Employee ID# 000076
Rigger/Driver 1

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

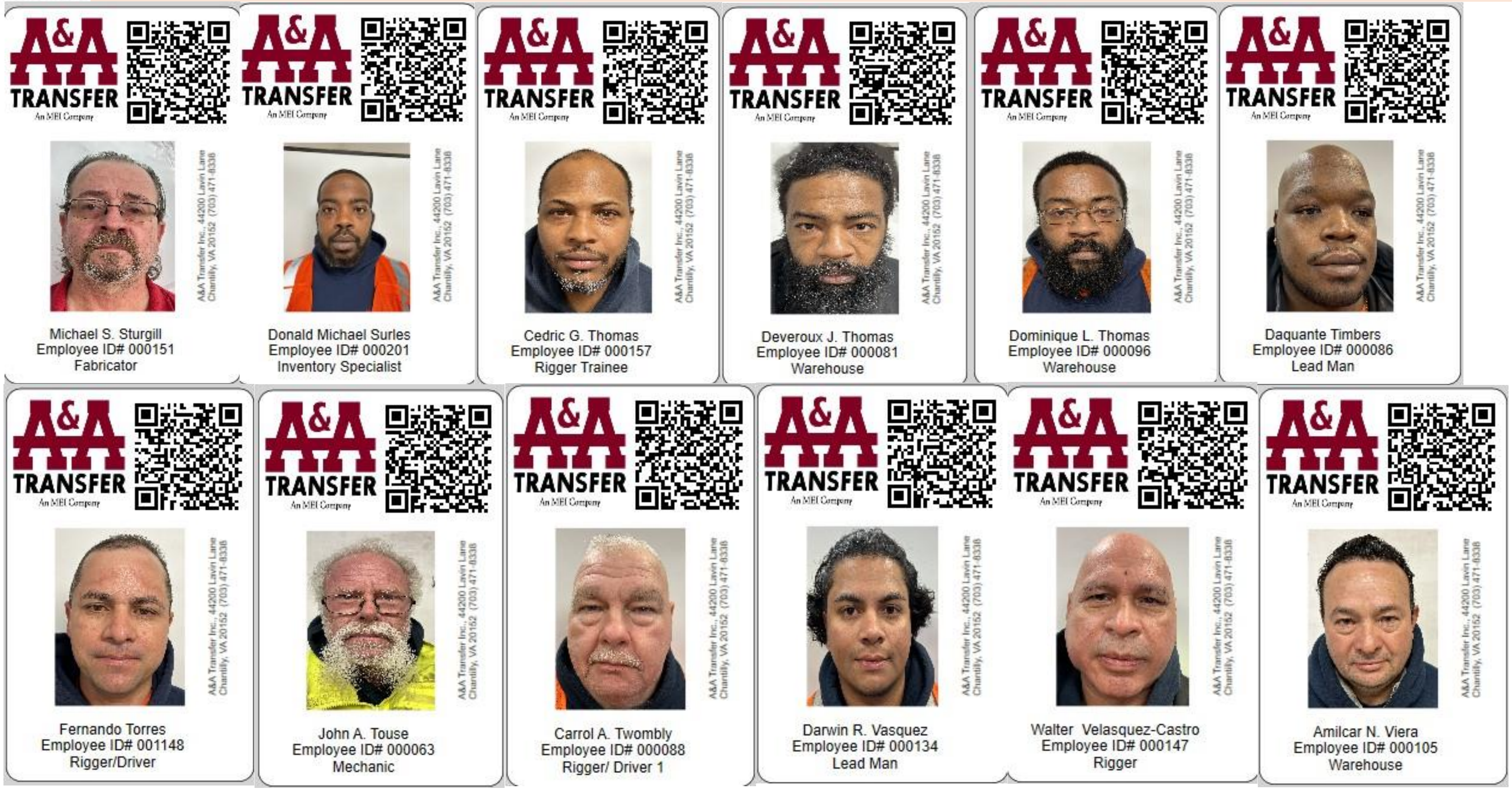
Zackary Rodgers
Employee ID# 000280
Lead Man

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Employee Qualification Cards FIELD



Employee Qualification Cards FIELD



Employee Qualification Cards FIELD

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Isaias J. Vigil Diaz
Employee ID# 000112
Rigger

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Darius Washington
Employee ID# 102081
Warehouse

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Darrel T. White
Employee ID# 001156
Mechanic

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Aspen Whiteside
Employee ID# 000172
Rigger

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Norman Wilkinson III
Employee ID# 101987
Rigger

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Enoch E. Wofford
Employee ID# 000267
Driver

TRANSFER
An MEI Company

A&A Transfer Inc., 44200 Lavin Lane
Chantilly, VA 20152 (703) 471-8336

Jose Zalaya Araujo
Employee ID# 102039
CDL Driver/Rigger 1

8 – Rigging Equipment Information

All rigging hardware, including shackles, utilized by A&A Transfer meets the requirement set forth in both 29 CFR 1926.251 (f)(1) & (2) and ANSI B30.26 26-1.5.1. A&A transfer regularly utilizes a variety of permissible shackles from different manufactures in the field. The shackles shown in this plan reflect a minimum WLL to be used in the field. Please do not hesitate to reach out to A&A Transfer's Dewey Snavely at 703-471-8338 with any questions or concerns regarding these statements.

liftex®
the world's best slings

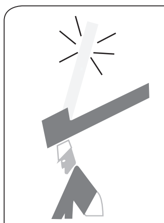
www.liftex.com
(800) 299-0900

ENDLESS ROUNDSLING CAPACITY CHART

	Part Number	Vertical	Choker	Basket	Minimum Length	Approx. Diameter	Approx. WT / FT
PURPLE	ENR1	2600	2100	5200	3’	.625”	0.3 LB
GREEN	ENR2	5300	4200	10600	3’	.875”	0.4 LB
YELLOW	ENR3	8400	6700	16800	3’	1.125”	0.5 LB
TAN	ENR4	10600	8500	21200	3’	1.125”	0.6 LB
RED	ENR5	13200	10600	26400	3’	1.375”	0.8 LB
WHITE	ENR6	16800	13400	33600	6’	1.375”	0.9 LB
BLUE	ENR7	21200	17000	42400	6’	1.625”	1.3 LB
ORANGE	ENR8	25000	20000	50000	6’	1.750”	1.6 LB
ORANGE	ENR9	31000	24800	62000	6’	2.125”	2.0 LB
ORANGE	ENR10	40000	32000	80000	6’	2.350”	2.6 LB
ORANGE	ENR11	53000	42400	106000	8’	3.150”	3.4 LB
ORANGE	ENR12	66000	52800	132000	8’	3.950”	4.3 LB
ORANGE	ENR13	90000	72000	180000	8’	4.800”	5.9 LB

**Before ordering slings that are going to be used in a chemically active environment, contact Liftex® Customer Service, to recommend the right sling for the right usage.*

Chemically Active Environments can affect the strength of webbing slings in varying degrees, ranging from little to total degradation.



WARNING

SLING FAILURE CAN CAUSE

DEATH OR INJURY

SLING FAILURE RESULTS FROM MISUSE, DAMAGE, AND EXCESSIVE WEAR

ADHERE TO INDUSTRY STANDARDS & REGULATIONS

CROSBY G209 & G213

TEL 01784 464447   FAX 01784 454788

SCREW PIN

G-209 S-209

Screw pin anchor shackles meet the performance requirements of Federal Specification RR-C-271D Type IVA, Grade A, Class 2, except for those provisions required of the contractor.

- Working Load Limit permanently shown on every shackle.
- Forged - Quenched and Tempered, with alloy pins.
- Capacities 1/3 thru 55 tons.
- Look for the Red Pin™ . . . the mark of genuine Crosby quality.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certification. Charges for proof testing and certification available when requested at the time of order.
- Hot Dip galvanized or Self Colored.
- Fatigue rated.

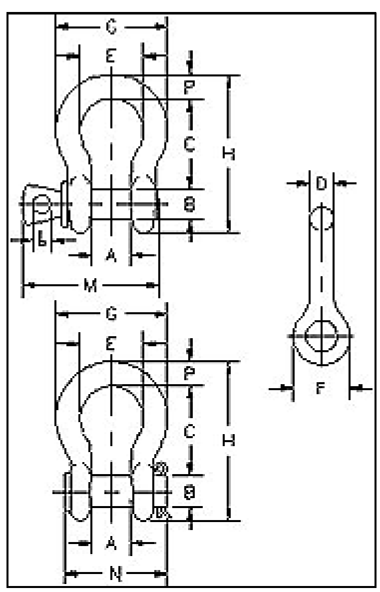
ROUND PIN

G-213 S-213

Round pin anchor shackles meet the performance requirements of Federal Specification RR-C-271D Type IVA, Grade A, Class 1, except for those provisions required of the contractor.

Nominal Size (in.)	Working Load Limit * (tons)	Stock No.				Weight Each (lbs.)	
		G-209 Galv.	S-209 S.C.	G-213 Galv.	S-213 S.C.	G-209 S-209	G-213 S-213
3/16	† 1/3	1018357	—	—	—	.06	—
1/4	1/2	1018375	1018384	1018017	1018026	.10	.13
5/16	3/4	1018393	1018400	1018035	1018044	.19	.18
3/8	1	1018419	1018428	1018053	1018062	.31	.29
7/16	1-1/2	1018437	1018446	1018071	1018080	.38	.38
1/2	2	1018455	1018464	1018099	1018106	.72	.71
5/8	3-1/4	1018473	1018482	1018115	1018124	1.37	1.50
3/4	4-3/4	1018491	1018507	1018133	1018142	2.35	2.32
7/8	6-1/2	1018516	1018525	1018151	1018160	3.62	3.49
1	8-1/2	1018534	1018543	1018179	1018188	5.03	5.00
1-1/8	9-1/2	1018552	1018561	1018197	1018204	7.41	6.97
1-1/4	12	1018570	1018589	1018213	1018222	9.50	9.75
1-3/8	13-1/2	1018598	1018605	1018231	1018240	13.53	13.25
1-1/2	17	1018614	1018623	1018259	1018268	17.20	17.25
1-3/4	25	1018632	1018641	1018277	1018286	27.78	29.46
2	35	1018650	1018669	1018295	1018302	45.00	45.75
2-1/2	†55	1018678	1018687	—	—	85.75	—

G-209



CAMPBELL® CHAIN AND FITTINGS

Bolt Type
Anchor Shackles

	Carbon Steel Galvanized		Alloy Steel Galvanized	
Size (in.)	*Cat.No.*	*WLL (Ton)*	*Cat.No.*	*WLL (Ton)*
1/4"	5390435	1/2		
3/8"	5390635	1	5390695	2
1/2"	5390835	2	5390895	3-1/4
5/8"	5391035	3-1/4	5391095	5
3/4"	5391235	4-3/4	5391295	7
7/8"	5391435	6-1/2	5391495	9 1/2
1"	5391635	8-1/2	5391695	12-1/2
1-1/8"	5391835	9-1/2		
1-1/4"	5392035	12	5392095	18
1-3/8"	5392235	13-1/2	5392295	21
1-1/2"	5392435	17	5392495	30
1-3/4"	5392835	25		
2"	5393235	35		
2-1/2"	5394035	55		

Screw Pin
Anchor Shackles

	Carbon Steel Painted Blue		Carbon Steel Galvanized		Alloy Steel Galvanized	
Size (in.)	*Cat.No.*	*WLL (Ton)*	*Cat.No.*	*WLL(Ton)*	*Cat.No.*	*WLL(Ton)*
3/16"			5410335	1/3		
1/4"	5410405	1/2	5410435	1/2		
5/16"	5410505	3/4	5410535	3/4		
3/8"	5410605	1	5410635	1	5410695	2
7/16"	5410705	1-1/2	5410735	1-1/2		
1/2"	5410805	2	5410835	2	5410895	3-1/3
5/8"	5411005	3-1/4	5411035	3-1/4	5411095	5
3/4"	5411205	4-3/4	5411235	4-3/4	5411295	7
7/8"	5411405	6-1/2	5411435	6-1/2	5411495	9 1/2
1"	5411605	8-1/2	5411635	8-1/2	5411695	12-1/2
1-1/8"	5411805	9-1/2	5411835	9-1/2		
1-1/4"	5412005	12	5412035	12	5412095	18
1-3/8"	5412205	13-1/2	5412235	13-1/2	5412295	21
1-1/2"	5412405	17	5412435	17	5412495	30
1-3/4"	5412805	25	5412835	25		
2"	5413205	35	5413235	35		
2-1/2"			5414035	55		

SUPER STRONG ANCHOR SHACKLES

MADE IN THE USA

WORKING LOAD LIMIT: 1/2 TO 55 TONS

CM Super Strong Shackles are carbon-type shackles with strength ratings that are 17 to 50% stronger than comparable-sized carbon shackles. As a result, these shackles are designed with a 6:1 design factor. Anchor shackles can be side loaded or used for multiple connections.

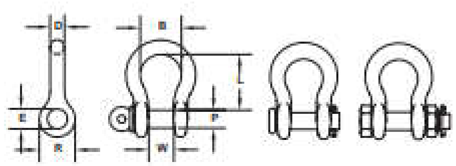
Screw Pin

BENEFITS & FEATURES

- Manufactured from technically advanced domestic (U.S.A.) micro alloy steel with optimal hardness for strength and ductility
- Shackles show major deformation before failure
- Working load limit and traceability codes shown as permanent markings on body
- All shackles have alloy quenched and tempered pins
- Available in sizes 3/16" to 2-1/2"
- Available finishes include powder coated, galvanized or self-colored
- Shackles meet dimensional requirements and exceed performance requirements of RR-C-271
- Special testing and certification is available if requested at the time of the order
- Note: Screw pin and bolt/nut/cotter shackles have a 6:1 design factor. 2-1/2" and all round pin shackles have a 5:1 design factor.
- Screw pin & bolt/nut/cotter shackles meet ASME B30.26

Round Pin

Bolt, Nut & Cotter



STYLES: Screw Pin, Round Pin, Bolt/Nut/Cotter
FINISHES: Self Colored, Galvanized, Orange Powder Coated

Size D (in.)	Working Load Limit (Ton)	Std. Pkg.	Weight (lbs.)	Product Code									Dimensions (in.)					
				Screw Pin			Round Pin			Bolt, Nut & Cotter								
				Self Colored	Galva-nized	Orange Powder Coated	Self Colored	Galva-nized	Orange Powder Coated	Self Colored	Galva-nized	Orange Powder Coated	P	E	W	R	L	B (min.)
3/16	1/2	50	0.06	–	M645G	–	–	M345G	–	–	–	–	0.25	0.29	0.38	0.57	0.88	0.58
1/4	3/4	50	0.12	M646	M646G	M646P	M346	M346G	M346P	M846	M846G	M846P	0.31	0.36	0.47	0.75	1.13	0.75
5/16	1	50	0.20	M647	M647G	M647P	M347	M347G	M347P	M847	M847G	M847P	0.38	0.45	0.53	0.84	1.25	0.81
3/8	1-1/2	50	0.30	M648	M648G	M648P	M348	M348G	M348P	M848	M848G	M848P	0.44	0.52	0.66	1.00	1.40	1.00
7/16	2	50	0.50	M649	M649G	M649P	M349	M349G	M349P	M849	M849G	M849P	0.50	0.58	0.72	1.15	1.69	1.19
1/2	3	50	0.75	M650	M650G	M650P	M350	M350G	M350P	M850	M850G	M850P	0.63	0.70	0.84	1.34	1.94	1.38
5/8	4-1/2	25	1.30	M651	M651G	M651P	M351	M351G	M351P	M851	M851G	M851P	0.75	0.83	1.06	1.66	2.41	1.63
3/4	6-1/2	10	2.30	M652	M652G	M652P	M352	M352G	M352P	M852	M852G	M852P	0.88	0.95	1.28	1.94	2.84	1.89
7/8	8-1/2	10	3.50	M653	M653G	M653P	M353	M353G	M353P	M853	M853G	M853P	1.00	1.09	1.44	2.14	3.31	2.06
1	10	5	5.00	M654	M654G	M654P	M354	M354G	M354P	M854	M854G	M854P	1.13	1.22	1.72	2.44	3.75	2.52
1-1/8	12	Bulk	7.00	M655	M655G	M655P	M355	M355G	M355P	M855	M855G	M855P	1.25	1.36	1.84	2.66	4.02	2.69
1-1/4	14	Bulk	9.50	M656	M656G	M656P	M356	M356G	M356P	M856	M856G	M856P	1.38	1.52	2.03	3.15	4.63	2.88
1-3/8	17	Bulk	12.50	M666	M666G	M666P	M366	M366G	M366P	M866	M866G	M866P	1.50	1.65	2.25	3.25	5.19	3.25
1-1/2	20	Bulk	17.20	M657	M657G	M657P	M357	M357G	M357P	M857	M857G	M857P	1.63	1.77	2.41	3.50	5.63	3.50
1-5/8	24	Bulk	23.90	M685	M685G	M685P	M385	M385G	M385P	M885	M885G	M885P	1.75	1.88	2.66	3.91	6.13	4.13
1-3/4	30	Bulk	27.70	M677	M677G	M677P	M377	M377G	M377P	M877	M877G	M877P	2.00	2.13	2.94	4.06	6.97	4.75
2	35	Bulk	39.00	M658	M658G	M658P	M358	M358G	M358P	M858	M858G	M858P	2.25	2.38	3.28	4.51	7.44	5.50
2-1/2	55	Bulk	90.90	–	–	–	–	–	–	MC860	MC860G	–	2.75	2.91	4.13	6.25	10.48	6.75

Green Pin® Bow Shackle SC

Standard bow shackle with screw collar pin

Highlights

- Screw pin for quick (dis)assembly
- Galvanization assures long-term durability
- Suitable for both one-leg and multi-leg systems
- Conforms to wide range of certifications (e.g. DNV)
- Superior stock availability of 99%

Description

The Green Pin® Bow Shackle SC is a bow shackle with a screw collar pin. The screw pin enables quick (dis)assembly which makes the shackle perfect for rigging activities in which assembly and disassembly occur relatively frequently. The Green Pin® Bow Shackle SC can be used for both one-leg and multi-leg systems. Due to the galvanization of the shackle its long-term durability is assured. The shackle's design reduces the wear of (wire) rope as the screw thread is hidden in a chamber. Particularly important for use offshore, the Green Pin® Bow Shackle SC conforms to a wide range of certifications from class societies such as DNV GL. The shackle is available in a range with a working load limit from 0.33 up to 55 ton.

Product details

Productcode	G-4161
Material	bow and pin high tensile steel, Grade 6, quenched and tempered
Safety factor	MBL equals 6 x WLL
Finish	hot dipped galvanized
Certification	**2.1 2.2 3.1 MTCa DNVGL-ST-0378 CE ABS PDA ABS MA**
Standard	EN 13889 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 2, Grade A, from 2 t and upward these shackles comply with ASME B30.26

Green Pin® Bow Shackle SC

Technical specifications in inches

Article code	A diameter bow (inch)	B diameter pin (inch)	C diameter eye (inch)	D width eye (inch)	E width inside (inch)	F length inside (inch)	G width bow (inch)	H length (inch)	I length bolt (inch)	J width (inch)	Net weight (LBS)
GPGHBB05	3/16	1/4	1/2	3/16	3/8	7/8	5/8	1 13/32	1 5/32	1 1/32	0.05
GPGHBB06	1/4	5/16	5/8	9/32	15/32	1 5/32	25/32	1 7/8	1 1/2	1 11/32	0.12
GPGHBB08	5/16	3/8	25/32	11/32	17/32	1 1/4	7/8	2 7/32	1 27/32	1 9/16	0.22
GPGHBB10	3/8	7/16	29/32	13/32	21/32	1 15/32	1 1/32	2 17/32	2 1/8	1 13/16	0.30
GPGHBB11	7/16	1/2	31/32	7/16	3/4	1 11/16	1 5/32	2 7/8	2 11/32	2	0.42
GPGHBB13	1/2	5/8	1 11/32	1/2	7/8	2	1 1/4	3 17/32	2 7/8	2 5/16	0.79
GPGHBB16	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 1/2	2 15/16	1.38
GPGHBB19	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 1/16	3 1/2	2.22
GPGHBB22	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	4 11/16	4 1/32	3.31
GPGHBB25	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 13/32	4 21/32	4.86
GPGHBB28	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 1/32	5 5/32	6.97
GPGHBB32	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	6 11/16	5 25/32	9.49

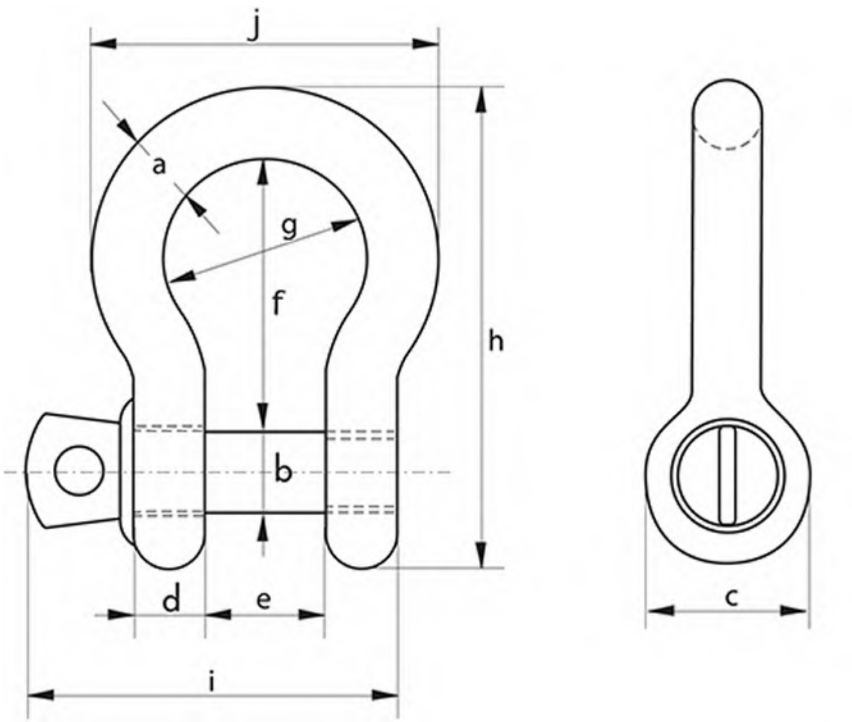
Green Pin® Bow Shackle SC

Technical specifications in inches

Article code	A diameter bow (inch)	B diameter pin (inch)	C diameter eye (inch)	D width eye (inch)	E width inside (inch)	F length inside (inch)	G width bow (inch)	H length (inch)	I length bolt (inch)	J width (inch)	Net weight (LBS)
GPGHBB35	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 5/16	6 3/8	12.3
GPGHBB38	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	8	6 7/8	16.4
GPGHBB45	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	27.7
GPGHBB50	2	2 1/4	4 13/32	1 31/32	3 9/32	7 3/4	5 7/16	13 1/16	10 23/32	9 3/8	37.8
GPGHBB57	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	8 3/4	6 5/16	14 7/8	12 7/32	10 25/32	58.0
GPGHBB65	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	10 1/4	7 3/32	17 1/16	13 17/32	12 7/32	82.9

Green Pin® Bow Shackle SC

Technical drawing



9 - Inclement Weather Operation

Standard Operating Procedure

Title: **Crane Inclement Weather Operation (High Winds and Thunderstorms)**	Version Number: 01	Effective Date: **02/01/2021**	Page 1 of 2

1 Purpose

Standard operating procedure for crane operations under high wind conditions and thunderstorms.

2 Scope

2.1 Crane operating procedures for jobsites when wind conditions sustained, or gusts exceed 20mph.

2.2 Crane operating procedures for jobsite when a thunderstorm is within a 10 mile radius of operation.

3 Definitions/Acronyms

3.1 ***Hoisting*** is the act of raising, lowering or otherwise moving a load in the air with equipment.

3.2 ***Qualified Person*** means a person who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, successfully demonstrated the ability to solve/resolve problems relating to the subject matter, the work, or the project. (1926.1401)

4 Procedures

4.1 High Winds: If high winds make the operation of the crane unsafe then the operator will boom down until it is safe to operate.

4.2 Wind speeds (sustained or gusts) of 20mph, operations are stopped. Conditions are discussed with qualified person (Operator) and the lift supervisor and site supervisor as whether to proceed or stop with operations.

4.3 Manufacturer's recommendations are to be followed at wind speeds (sustained or gusts) not exceeding 30mph at the suspended load.

- Hoisting man-cage be performed in a slow, controlled, cautious manner, with no sudden movements of the equipment or man-cage.
- When windspeeds or gusts exceed 25mph at the man-cage, all operations involving the hoisting of personnel shall stop, until such time is safe to operate.
- For all other material hoisted by the crane operator, operations will stop at windspeeds exceeding 30 mph unless the crane manufacturer's operations manual dictates a lower threshold.

- Qualified person (Operator) shall decide for loads with large surface areas relative to size whether the load can be safely hoisted regardless of the wind criteria above.

4.4 Thunderstorms: When using a crane, check the local NOAA weather reports first to see if there's a chance of a thunderstorm. If there's a chance of thunder and lightning could be in the area, you should plan to use the crane on a different day or work around the timing of the storm.

- Stop using crane when a thunderstorm is within 10 miles of operation.
- Shutdown crane operation if you see lightning or hear thunder. Those are two signs that indicate thunderstorm is close enough to potentially put jobsite in danger.
- Wait 30 minutes after thunderstorm has passed to continue crane operation. Prior to resuming operation, visually inspect crane to determine if any damage may have occurred during the storm. If any damage has occurred do not use equipment until further inspection and repairs are completed.

5 References

- 29 CFR 1926 OSHA: Subpart CC – Cranes Derricks in Construction 1926.1400
- Crane Manufacturer's Operating Manual
- Local NOAA Weather Reports

END OF DOCUMENT